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Towards Learning Monocular 3D Object Localization From 2D Labels Using the Physical Laws of Motion

利用运动物理定律，仅从二维标注学习单目三维目标定位

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Abstract
摘要

We present a novel method for precise 3D object localization in single images from a single calibrated camera using only 2D labels. No expensive 3D labels are needed. Thus, instead of using 3D labels, our model is trained with easy-to-annotate 2D labels along with the physical knowledge of the object's motion. Given this information, the model can infer the latent third dimension, even though it has never seen this information during training. Our method is evaluated on both synthetic and real-world datasets, and we are able to achieve a mean distance error of just 6 cm in our experiments on real data. The results indicate the method's potential as a step towards learning 3D object location estimation, where collecting 3D data for training is not feasible.

我们提出一种精确的三维目标定位方法：输入单台已标定相机拍摄的单幅图像，**训练时仅需二维标注**，无需昂贵的三维标注。模型结合容易获取的二维标注与物体运动的物理知识进行训练，因而能够推断未观测到的第三维，即深度，即使训练时从未接触该维度的真值信息。我们在合成数据集和真实数据集上评估该方法，真实数据实验的平均距离误差仅为 6 cm。结果表明，对于难以收集三维训练数据的情形，本方法为学习三维目标定位提供了一条可行途径。

1 Introduction

1 简介

The 3D location of objects is crucial in many application domains, such as in robotics or in the performance analysis of athletes in sports tournaments.

对象的 3D 位置在许多应用领域中至关重要，例如机器人技术或体育比赛中运动员的表现分析。

Especially sports broadcasting companies are greatly interested in obtaining 3D information about an object’ s location in a fixed environment [22] , for example the position of the ball in soccer, basketball, tennis, or squash.

尤其是体育转播公司对获取固定环境中物体位置的 3D 信息非常感兴趣 [22]，例如足球、篮球、网球或壁球中球的位置。

Traditional technologies like Hawk-Eye [13] and View 4D [32] often rely on triangulation techniques [21 , 23 , 26] for the calculation of the ball’ s position.

Hawk-Eye [13] 和 View 4D [32] 等传统技术通常依赖三角测量技术 [21,23,26] 来计算球的位置。

A major limitation of these methods is the need for expensive hardware (e.g. multiple synchronized cameras), consequently preventing their application in low budget and amateur sports where only a single camera is used to record the game.

这些方法的一个主要限制是需要昂贵的硬件（例如多个同步摄像机），从而阻碍了它们在仅使用单个摄像机来记录比赛的低预算和业余运动中的应用。

In contrast to these methods, a neural network can be trained to predict the 3D position of the ball even if the game is recorded with just a single camera.

与这些方法相比，即使仅用单个摄像机记录比赛，也可以训练神经网络来预测球的 3D 位置。

However, training such a network typically requires 3D ground truth data.

然而，训练这样的网络通常需要 3D 真值数据。

Although ground truth data may be readily available for some disciplines such as soccer, it is often lacking in lower-budget sports like squash.

尽管足球等某些项目的真值数据可能很容易获得，但壁球等低成本运动往往缺乏真值数据。

Consequently, there is a need for a method that can perform ball localization without relying on 3D ground truth data as supervision.

因此，需要一种能够在不依赖3D真值数据作为监督的情况下执行球定位的方法。

In this paper, we present a novel method for training a neural network to perform 3D localization that does not need 3D ground truth labels as supervision at all.

本文提出一种训练神经网络进行三维定位的新方法，**完全不需要三维真值标签**作为监督。

Instead, only 2D image labels of the position of a desired object are needed, which can be easily obtained through methods such as object detection or manual annotation by clicking on the object’ s center in an image.

只需提供目标位置的二维图像标注；这些标注易于获取，例如可通过目标检测获得，也可在图像中点击目标中心进行人工标注。

Our model is trained using video clips, leveraging the physical laws of motion to enable the network to infer the latent third dimension.

模型通过视频片段训练，并利用物体的运动规律推断未观测到的第三维，即深度。

A key advantage of our method is its flexibility in inference.

我们的方法的一个关键优势是其推理的灵活性。

While videos are needed during training, our trained model can also be applied to single images during inference.

虽然训练期间需要视频，但我们训练的模型也可以在推理期间应用于单个图像。

This ensures that our network can even be applied in situations where the object’ s motion cannot be precisely described by physics (e.g. due to human interaction) or where the physical parameters are not known completely (e.g. if a sudden wind is affecting the ball).

这确保了我们的网络甚至可以应用于物理无法精确描述物体运动的情况（例如由于人类交互）或物理参数不完全已知的情况（例如突然的风影响球）。

While methods processing multiple frame will likely fail in those situations, our method is robust to such changes.

虽然处理多帧的方法在这些情况下可能会失败，但我们的方法对此类变化具有鲁棒性。

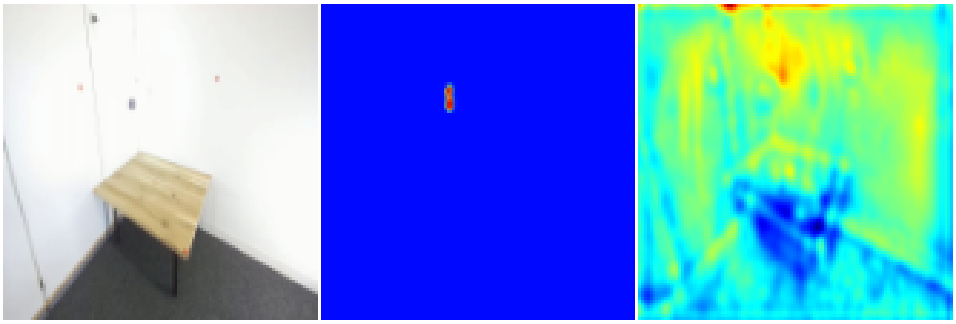


Figure 1 : Predictions of the Position Estimation Network .

图 1：位置估计网络的预测。

The left image is the input to the network, the middle image shows the predicted heatmap, and the right image shows the predicted depthmap.

左图是网络的输入，中图显示预测的热图，右图显示预测的深度图。

By combining the heatmap and depthmap, the 3D position of the ball can be calculated.

通过结合热图和深度图，可以计算出球的 3D 位置。

The primary focus of this paper is on training our neural network on video sequences of the ball’ s movement between contacts with players.

本文的主要重点是根据球与球员接触之间运动的视频序列来训练我们的神经网络。

During this period, the ball remains unaffected by non-deterministic forces, enabling us to effectively utilize the physical laws of motion to describe its trajectory in the training videos.

在此期间，球不受非确定性力的影响，使我们能够有效地利用运动的物理定律来描述其在训练视频中的轨迹。

Importantly, we are able to describe the ball bouncing off walls and the floor, which is e.g. important in various disciplines like tennis or squash.

我们还能够描述球在墙壁和地面上的反弹，这对网球、壁球等运动尤为重要。

Since our method is not only limited to sport’ s applications, we study videos of a moving ball in a more general context.

由于我们的方法不仅限于体育应用，因此我们在更广泛的背景下研究移动球的视频。

The main contributions of this paper are:

本文的主要贡献是：

- We propose a general method to train a neural network for 3D object localization in single images without the need of 3D labels.

  我们提出了一种通用方法来训练神经网络以在单个图像中进行 三维目标定位，而无需 3D 标签。

- We create a synthetic and a real-world dataset with 3D ground truths for the evaluation of our method.

  我们构建了具有三维真值的合成数据集和真实数据集，用于评估本方法。

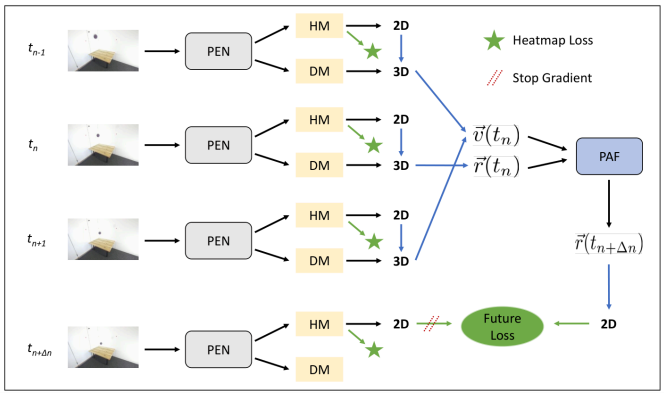
  The datasets will be published in order to ensure reproducibility and to encourage further research in this field.

  这些数据集将被发布，以确保可重复性并鼓励该领域的进一步研究。

-

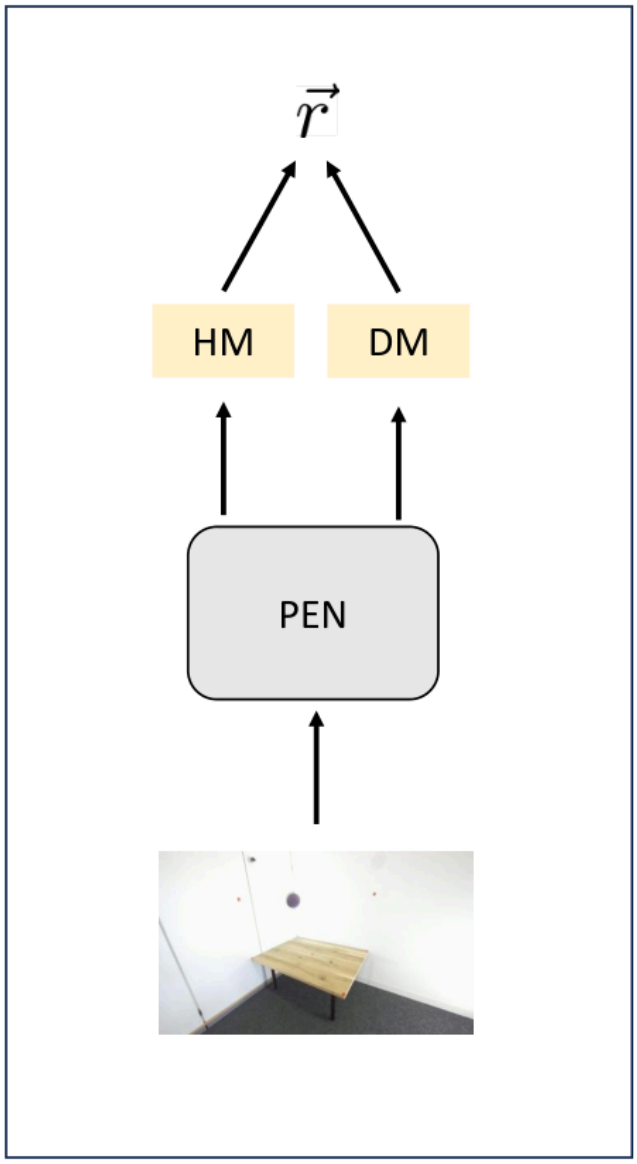
We prove experimentally that our model is able to accurately predict the 3D position of the ball and discuss the experimental results thoroughly.

我们通过实验证明我们的模型能够准确预测球的 3D 位置，并彻底讨论实验结果。



(a) Training with videos using explicit physical knowledge.

(a) 使用明确的物理知识进行视频训练。



(b) Inference with single images.

(b) 单图像推理。

No explicit physical knowledge necessary.

不需要明确的物理知识。

Figure 2 : Training and inference pipeline.

图 2：训练和推理流程。

The Position Estimation Network (PEN) estimates the heatmap (HM) and depthmap (DM) for each input image, from which the 2D image coordinates and camera depth are extracted.

位置估计网络 (PEN) 估计每个输入图像的热图 (HM) 和深度图 (DM)，从中提取 2D 图像坐标和相机深度。

Using the intrinsic camera matrix, we compute the 3D camera coordinates from the image coordinates and camera depth.

使用相机内参矩阵，我们根据图像坐标和相机深度计算 3D 相机坐标。

With the extrinsic camera matrix, the world coordinates $\vec{r}$ are calculated.

利用相机外参矩阵，计算出世界坐标$\vec{r}$。

During training, we apply the PEN to multiple coherent frames for the estimation of the velocity $\vec{v}$.

训练时，将 PEN 应用于时间上连续的多帧图像，以估计速度 $\vec{v}$。

The Physics Aware Forecast Module (PAF) utilizes these initial conditions to calculate the world coordinates at a later time $t_{n+\Delta n}$ by solving the differential equations of motion.

物理感知预测模块 (PAF) 利用这些初始条件通过求解运动微分方程来计算稍后的世界坐标$t_{n+\Delta n}$。

These coordinates are then projected back to image coordinates and the future loss is calculated.

然后将这些坐标投影回图像坐标并计算未来预测损失。

The heatmap loss is calculated by comparing the predicted and ground truth heatmaps at all time steps.

热图损失是通过比较所有时间步的预测热图和真值热图来计算的。

During inference, the PEN is applied to single images.

在推理过程中，PEN 应用于单个图像。

Thus, it is robust to incalculable situations like sudden wind or human interventions.

因此，它对于不可估量的情况（如突然的风或人为干预）具有鲁棒性。

Blue arrows indicate the use of camera matrices, and dotted red lines indicate detached gradients.

蓝色箭头表示使用相机矩阵的坐标变换，红色虚线表示停止梯度传播的位置。

2 Related Work

2 相关工作

Pose estimation is widely used in many sport disciplines, and it is of great importance in athletics [20] , table tennis [16] , swimming [9] , and soccer [2] .

姿态估计广泛应用于许多运动学科，在田径运动[20]、乒乓球[16]、游泳[9]和足球[2]中非常重要。

In this paper we estimate the position of an object instead of a human pose.

本文估计的是物体位置，而非人体姿态。

Nevertheless, our architecture is inspired by methods commonly used in human pose estimation, since we use 2D heatmaps to encode the object’ s position.

尽管如此，我们的架构受到了人体姿态估计中常用方法的启发，因为我们使用 2D 热图来编码对象的位置。

In addition to human pose estimation, accurate ball localization is crucial in the analysis of many sport disciplines.

除了人体姿态估计之外，准确的球定位对于许多运动学科的分析也至关重要。

Several methods have been developed to calculate physical reasonable 3D positions of the ball from the sequence of related 2D positions.

已经开发了几种方法来根据相关 2D 位置序列计算球的物理合理 3D 位置。

For instance, [18] localize a badminton ball using videos, [5] reconstruct the 3D trajectory of a basketball from a sequence of 2D positions and [6] calculate the trajectory of a volleyball.

例如，[18]使用视频定位羽毛球，[5]从一系列 2D 位置重建篮球的 3D 轨迹，[6]计算排球的轨迹。

However, these methods rely on fitting the ball’ s trajectory to physically calculated paths and, thus, are only applicable to video sequences exhibiting perfect ballistic trajectories.

然而，这些方法依赖于将球的轨迹拟合到物理计算的路径，因此仅适用于展示完美弹道轨迹的视频序列。

They cannot handle situations where the ball is subject to non-deterministic forces (e.g., human actions) or when the physical parameters change (e.g., the introduction of strong winds or changes in drag coefficients due to wear and tear).

它们无法处理球受到非确定性力（例如，人类行为）或物理参数发生变化（例如，引入强风或由于磨损而导致阻力系数变化）的情况。

Moreover, these methods do not account for ball bouncing.

此外，这些方法没有考虑球的弹跳。

In contrast to these methods, we are also able to describe more intricate physical situations like a ball bouncing off the floor.

与这些方法相反，我们还能够描述更复杂的物理情况，例如球从地面上弹起。

Furthermore, since our method can be applied to single images during inference, it remains effective even when the ball’s behavior is non-deterministic or when precise knowledge of physical parameters is lacking.

此外，由于我们的方法可以在推理过程中应用于单个图像，因此即使球的行为不确定或缺乏物理参数的精确知识，它仍然有效。

Thus, our method overcomes many limitations of previous approaches.

因此，我们的方法克服了以前方法的许多限制。

Other studies like [31 , 30] estimate the 3D position of a basketball in single images by measuring the diameter of the ball in the images.

其他研究如 [31, 30] 通过测量图像中球的直径来估计单个图像中篮球的 3D 位置。

Similarly, [4] utilizes the ball diameter to predict the 3D position of a table tennis ball.

类似地，[4]利用球直径来预测乒乓球的3D位置。

Nevertheless, for these methods additional relatively costly segmentation labels of the ball are needed.

然而，对于这些方法来说，需要额外的相对昂贵的球分割标签。

In contrast, our method only needs less expensive 2D image-coordinate annotations of the ball either from manual or automatic annotations.

相比之下，我们的方法只需要通过手动或自动标注对球进行较便宜的 2D 图像坐标标注。

To the best of our knowledge, we are the first to develop a method for localizing the ball’s 3D center coordinates in single images using only 2D labels together with knowledge of the physical motion.

据我们所知，我们是第一个开发出一种仅使用 2D 标签以及物理运动知识来在单个图像中定位球的 3D 中心坐标的方法。

Even though we focus on the 3D localization of a ball, it is worth noting that our method can be applied to other objects as well.

虽然本文重点研究球的三维定位，但该方法也适用于其他物体。

As we utilize the physical knowledge of the moving ball for training our neural network, its physical motion has to be described in a differentiable manner.

当我们利用移动球的物理知识来训练我们的神经网络时，它的物理运动必须以可微分的方式描述。

While the motion can be described by an analytic function for many toy problems, the physical equations of motion have to be solved numerically for most real life applications.

对于许多简化问题，运动可用解析函数描述；但在多数实际应用中，运动方程需要数值求解。

We implement the differential equations using the framework of Neural Ordinary Differential Equations (NDEs) [7] in order to be able to calculate the gradient of the numerical solution.

我们使用神经常微分方程（NDE）[7]的框架来实现微分方程，以便能够计算数值解的梯度。

Our method furthermore resembles Hamiltonian Neural Networks (HNNs) and Lagrangian Neural Networks (LNNs) [11] .

我们的方法还类似于哈密顿神经网络（HNN）和拉格朗日神经网络（LNN）[11]。

However, in contrast to HNNs and LNNs, we learn the ball’s 3D coordinates given a fixed Hamilton function instead of learning the Hamilton or Lagrange function given fixed data.

不同于 HNN 和 LNN 在给定数据下学习哈密顿函数或拉格朗日函数，本方法固定哈密顿函数，学习球的三维坐标。

3 Method

3 方法

3.1 Overview

3.1 概述

Our method consists of two main modules: A Position Estimation Network (PEN) and a Physics Aware Forecast Module (PAF).

我们的方法由两个主要模块组成：位置估计网络（PEN）和物理感知预测模块（PAF）。

The PEN is a neural network that takes a single image as input and generates a heatmap as well as a depthmap as output.

PEN 是一种神经网络，它将单个图像作为输入并生成热图和深度图作为输出。

By extracting the 2D image coordinates from the heatmap and using the depth information from the depthmap along with the intrinsic and extrinsic camera calibration matrix, we calculate the camera and world coordinates of the ball.

通过从热图中提取 2D 图像坐标，并使用深度图中的深度信息以及相机内参和外参标定矩阵，我们计算球的相机和世界坐标。

The PEN is applied to three successive images at time t_{n-1}, t_n, t_{n+1} to obtain the ball’s world-coordinates for each image.

PEN 在时间t_{n-1}, t_n, t_{n+1}处应用于三个连续图像，以获得每个图像的球的世界坐标。

Using the coordinates at time t_{n-1} and t_{n+1}, we estimate the ball’s velocity at time t_n.

使用t_{n-1}和t_{n+1}时刻的坐标，我们估计t_n时刻球的速度。

Given the ball’s world coordinates and velocity at time t_n, the PAF calculates the world coordinates at a later time $t_{n+\Delta n}$ by solving the differential equations of motion.

给定球在t_n时刻的世界坐标和速度，PAF 通过求解运动微分方程来计算稍后$t_{n+\Delta n}$时刻的世界坐标。

We implement two losses for the training of the PEN: The heatmap loss and the future loss .

我们在 PEN 的训练中实现了两种损失：热图损失和未来预测损失。

The heatmap loss compares the predicted and ground-truth heatmaps to teach the network the ball’s 2D image coordinates.

热图损失比较预测热图和真值热图，以向网络传授球的 2D 图像坐标。

The future loss ensures that the PEN learns the correct camera depth by projecting the PAF’s predicted coordinates at time $t_{n+\Delta n}$ to the 2D image coordinates for the frame at $t_{n+\Delta n}$.

未来预测损失确保 PEN 通过将时间$t_{n+\Delta n}$处的 PAF 预测坐标投影到$t_{n+\Delta n}$处帧的 2D 图像坐标来学习正确的相机深度。

An overview of our method is depicted in Figure 2 .

我们的方法的概述如图 2 所示。

We note that only the PEN is used during inference.

我们注意到，在推理过程中仅使用了 PEN。

3.2 Position Estimation Network (PEN)

3.2 位置估计网络（PEN）

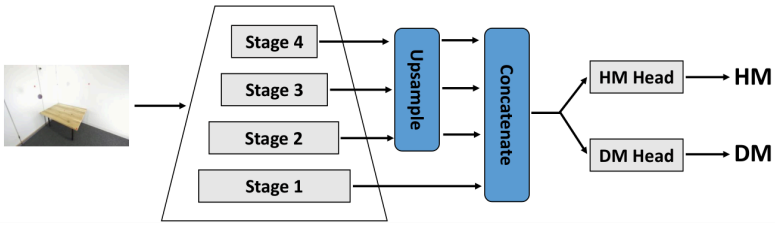


Figure 3 : Position Estimation Network (PEN) architecture for generating both heatmap (HM) and depthmap (DM) from an input image.

图 3：位置估计网络 (PEN) 架构，用于从输入图像生成热图 (HM) 和深度图 (DM)。

Features from the backbone network are concatenated and passed through the two heads.

来自主干网络络的特征被连接并通过两个头传递。

One generates a heatmap and the other produces a depthmap.

一个生成热图，另一个生成深度图。

The PEN takes an image $I \in \mathbb{R}^{C \times H \times W}$ as input with C , H and W being the number of channels, height and width of the image.

PEN 以图像$I \in \mathbb{R}^{C \times H \times W}$作为输入，$C$、$H$和$W$分别为图像的通道数、高度和宽度。

Using a backbone architecture like ResNet [14] or ConvNeXt [19 , 34] , the features after each stage are bilinearly upsampled to the shape of the first stage and, afterwards, they are concatenated.

使用像 ResNet [14] 或 ConvNeXt [19, 34] 这样的主干架构，每个阶段之后的特征都被双线性上采样到第一阶段的形状，然后将它们连接起来。

A heatmap head consisting of one 1×1 convolution, two 7×7 convolutions and a final 1×1 convolution is applied to obtain a heatmap $\tilde{H} \in \mathbb{R}^{H \times W}$ from the concatenated features.

应用由一个1×1卷积、两个7×7卷积和最后一个1×1卷积组成的热图头，从级联特征中获得热图$\tilde{H} \in \mathbb{R}^{H \times W}$。

Additionally, a separate depthmap head with the same architecture is applied to the concatenated features to calculate the depthmap $D \in \mathbb{R}^{H \times W}$.

此外，具有相同架构的单独深度图头被应用于级联特征以计算深度图$D \in \mathbb{R}^{H \times W}$。

We intentionally choose a simple architecture for the heads to demonstrate the effectiveness of our method without requiring extensive architecture modifications.

我们有意为头部选择一个简单的架构来证明我们方法的有效性，而不需要进行大量的架构修改。

The architecture of the PEN is depicted in Figure 3 .

PEN 的架构如图 3 所示。

The 2D image coordinates $\vec{\mathbf{r}}^{(I)} = \begin{pmatrix} x^{(I)} & y^{(I)} \end{pmatrix}^T$ are extracted from the heatmap with a 2D soft-argmax by weighting each pixel-position with its corresponding heatmap value.

通过使用 2D soft-argmax 对每个像素位置及其相应的热图值进行加权，从热图中提取 2D 图像坐标$\vec{\mathbf{r}}^{(I)} = \begin{pmatrix} x^{(I)} & y^{(I)} \end{pmatrix}^T$。

Furthermore, the camera depth $z^{(C)}$ is calculated as the sum over all values of the depthmap weighted with the corresponding values of the 2D softmax of the heatmap.

此外，相机深度$z^{(C)}$计算为深度图的所有值与热图的 2D softmax 的相应值加权的总和。

Consequently, the coordinates are calculated from the heatmap $\tilde{H}$ and the depthmap $\tilde{D}$ as

因此，根据热图$\tilde{H}$和深度图$\tilde{D}$计算坐标：

$$
\begin{aligned}
x^{(I)} &= \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} w \cdot \left(\operatorname{softmax}\left(\beta \cdot \widetilde{H}\right)\right)_{h,w} \\
y^{(I)} &= \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} h \cdot \left(\operatorname{softmax}\left(\beta \cdot \widetilde{H}\right)\right)_{h,w} \\
z^{(C)} &= \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} \widetilde{D}_{h,w} \cdot \left(\operatorname{softmax}\left(\beta \cdot \widetilde{H}\right)\right)_{h,w}
\end{aligned}
\tag{1}
$$

with the constant factor $\beta = 10$ to obtain sharp softmax probabilities and the pixel indices h and w .

其中常数 $\beta = 10$ 用于使 softmax 概率分布更集中，h 和 w 为像素索引。

3.3 Coordinate Transformations

3.3 坐标变换

We assume that the intrinsic camera matrix $\mathrm{M}_{\mathrm{int}}$ and the extrinsic camera matrix $\mathrm{M}_{\mathrm{ext}}$ are known, and we discuss in the supplementary material in more detail how the camera matrices can be obtained.

我们假设相机内参矩阵$\mathrm{M}_{\mathrm{int}}$和相机外参矩阵$\mathrm{M}_{\mathrm{ext}}$已知，并且我们在补充材料中更详细地讨论如何获得相机矩阵。

We use homogeneous coordinates and transform world coordinates into camera coordinates with the extrinsic matrix $\mathrm{M}_{\mathrm{ext}} \in \mathbb{R}^{4\times 4}$ and camera coordinates into image coordinates with the intrinsic matrix $\mathrm{M}_{\mathrm{int}} \in \mathbb{R}^{3\times 3}$.

我们使用齐次坐标，通过外参矩阵$\mathrm{M}_{\mathrm{ext}} \in \mathbb{R}^{4\times 4}$将世界坐标转换为相机坐标，并通过内参矩阵$\mathrm{M}_{\mathrm{int}} \in \mathbb{R}^{3\times 3}$将相机坐标转换为图像坐标。

Since these transformations are invertible we calculate the camera-coordinates $\vec{\mathbf{r}}^{(C)} = \begin{pmatrix} x^{(C)} & y^{(C)} & z^{(C)} \end{pmatrix}^T$ and the world-coordinates $\vec{\mathbf{r}}^{(W)} = \begin{pmatrix} x^{(W)} & y^{(W)} & z^{(W)} \end{pmatrix}^T$ given the image coordinates $\overrightarrow{\mathbf{r}^{(I)}}$ as well as the camera depth $z^{(C)}$ using the following formulas:

由于这些变换是可逆的，我们使用以下公式计算给定图像坐标$\overrightarrow{\mathbf{r}^{(I)}}$的相机坐标$\vec{\mathbf{r}}^{(C)} = \begin{pmatrix} x^{(C)} & y^{(C)} & z^{(C)} \end{pmatrix}^T$和世界坐标$\vec{\mathbf{r}}^{(W)} = \begin{pmatrix} x^{(W)} & y^{(W)} & z^{(W)} \end{pmatrix}^T$以及相机深度$z^{(C)}$:

$$
\begin{pmatrix} x^{(C)} \\ y^{(C)} \\ z^{(C)} \end{pmatrix} = \mathrm{M}_{\mathrm{int}}^{-1} \cdot \begin{pmatrix} x^{(I)} z^{(C)} \\ y^{(I)} z^{(C)} \\ z^{(C)} \end{pmatrix}, \quad \begin{pmatrix} x^{(W)} \\ y^{(W)} \\ z^{(W)} \\ 1 \end{pmatrix} = \mathrm{M}_{\mathrm{ext}}^{-1} \cdot \begin{pmatrix} x^{(C)} \\ y^{(C)} \\ z^{(C)} \\ 1 \end{pmatrix}.
\tag{2}
$$

Furthermore, the velocity $\vec{\mathbf{v}}(t_n)$ at time t_n is calculated from the ball’ s world coordinates at time t_{n-1} and t_{n+1} using the symmetric finite difference quotient as

此外，根据时间t_{n-1}和t_{n+1}时球的世界坐标，使用对称有限差分商计算出t_n时间的速度$\vec{\mathbf{v}}(t_n)$:

$$
\vec{\mathbf{v}}(t_n) = \frac{\vec{\mathbf{r}}^{(W)}(t_{n+1}) - \vec{\mathbf{r}}^{(W)}(t_{n-1})}{t_{n+1} - t_{n-1}}.
\tag{3}
$$

3.4 Physics Aware Forecasting Module (PAF)

3.4 物理感知预测模块（PAF）

Given the ball’ s world coordinates $\vec{\mathbf{r}}^{(W)}(t_n)$ and velocity $\vec{\mathbf{v}}^{(W)}(t_n)$ at time t_n as initial conditions, the PAF calculates the ball’ s coordinates at a later time $t_{n+\Delta n}$ by solving the differential equations of motion.

以t_n时刻球的世界坐标$\vec{\mathbf{r}}^{(W)}(t_n)$和速度$\vec{\mathbf{v}}^{(W)}(t_n)$作为初始条件，PAF通过求解运动微分方程计算出稍后$t_{n+\Delta n}$时刻球的坐标。

While the ball’ s motion can be expressed as an analytical function for many simple toy problems, it is not always possible to find an analytical solution for more intricate situations.

虽然球的运动可以表示为许多简单简化问题的解析函数，但并不总是能够找到更复杂情况的解析解。

Since we want to describe a bouncing ball in an arbitrary environment, we solve the differential equations of motion numerically.

由于我们想要描述任意环境中的弹跳球，因此我们对运动微分方程进行数值求解。

By using the framework of Neural Differential Equations [7] we are able to calculate the gradient of the numerical solution that is needed for the backpropagation algorithm.

借助神经微分方程框架 [7]，我们可以计算数值解的梯度，以进行反向传播。

We adopt the Hamilton formalism to describe the motion of the ball.

我们采用哈密顿力学描述球的运动。

A physical system can be described according to [12] using the Hamilton function

物理系统可以根据[12]使用哈密顿函数来描述

$$\mathcal{H} = \frac{\left|\vec{\mathbf{p}}\right|^2}{2m} + V(\vec{\mathbf{r}}^{(W)}) \tag{4}$$

with the ball’ s mass m , the momentum $\vec{\mathbf{p}} = m\vec{\mathbf{v}}$ (for euclidean coordinates) and the potential $V(\vec{\mathbf{r}}^{(W)})$, which describes the physical behavior of the ball.

球的质量m、动量$\vec{\mathbf{p}} = m\vec{\mathbf{v}}$（对于欧几里德坐标）和势$V(\vec{\mathbf{r}}^{(W)})$，描述了球的物理行为。

According to the Hamilton formalism [12] , the equations of motion are derived from the Hamilton function as

根据哈密顿形式[12]，运动方程由哈密顿函数导出为

$$\frac{d}{\mathrm{dt}}\vec{\mathbf{r}}^{(W)} = \frac{1}{m}\frac{d}{d\vec{\mathbf{v}}}\mathcal{H} \ , \qquad m\frac{d}{\mathrm{dt}}\vec{\mathbf{v}} = -\frac{d}{d\vec{\mathbf{r}}^{(W)}}\mathcal{H} \tag{5}$$

and by solving these differential equations given the initial ball position and velocity we get the ball’ s position $\vec{\mathbf{r}}^{(W)}(t)$ at time t .

给定球的初始位置和速度，通过求解这些微分方程，我们可以得到球在时间t的位置$\vec{\mathbf{r}}^{(W)}(t)$。

Consequently, the equations 5 are solved in the PAF to obtain the ball’ s position at a later time.

因此，在 PAF 中求解方程 5，以获得稍后球的位置。

The potential $V(\overrightarrow{\mathbf{r}^{(W)}})$ defines the physical behavior of the ball and, thus, we have to model it accordingly to ensure the correct prediction of the ball’ s movement.

势$V(\overrightarrow{\mathbf{r}^{(W)}})$定义了球的物理行为，因此，我们必须对其进行相应的建模，以确保正确预测球的运动。

The potential of a ball in free fall above the floor at $z^{(W)} = 0$ is described as

对于位于 $z^{(W)} = 0$ 处地面上方、做自由落体运动的球，其势能为

$$V_G(\vec{\mathbf{r}}^{(W)}) = m \cdot g \cdot \mathrm{ReLU}(z^{(W)}) \tag{6}$$

with g being the gravitational constant.

其中 g 为重力加速度。

Additionally, we also want to describe the bouncing off the floor.

此外，我们还想描述从地面上弹起的情况。

Ideally, the potential at the floor position is described as an infinite potential barrier.

理想情况下，地面位置对应一个无限高的势垒。

However, it is sufficient to approximate the infinite barrier as a fast increasing potential.

然而，将无限势垒近似为快速增加的势就足够了。

Therefore, we model the potential at the floor position as

因此，地面位置的势能建模为

$$V_F(\vec{\mathbf{r}}^{(W)}) = m \cdot c \cdot \mathrm{ReLU}(-z^{(W)}) \tag{7}$$

with $c = 1000\ \frac{J}{m \cdot kg}$ being a large constant.

其中$c = 1000\ \frac{J}{m \cdot kg}$是一个大常数。

We note that the exact value of c is not important as long as it is significantly larger than g .

我们注意到，c的确切值并不重要，只要它显著大于g即可。

In some environments, we also have to include the bouncing off the walls or various obstacles.

在某些环境中，我们还必须考虑墙壁或各种障碍物的弹跳。

These can be defined similarly and are explained in the supplementary material for our specific datasets in more detail.

这些可以类似地定义，并在我们特定数据集的补充材料中进行更详细的解释。

As a result, the overall potential of a moving ball is described as

于是，运动球的总势能为

$$V(\vec{\mathbf{r}}^{(W)}) = V_G(\vec{\mathbf{r}}^{(W)}) + V_F(\vec{\mathbf{r}}^{(W)}) + V_W(\vec{\mathbf{r}}^{(W)}) + V_O(\vec{\mathbf{r}}^{(W)}) \tag{8}$$

with $V_W(\vec{\mathbf{r}}^{(W)})$ describing the bouncing off the walls and $V_O(\vec{\mathbf{r}}^{(W)})$ describing the bouncing off obstacles.

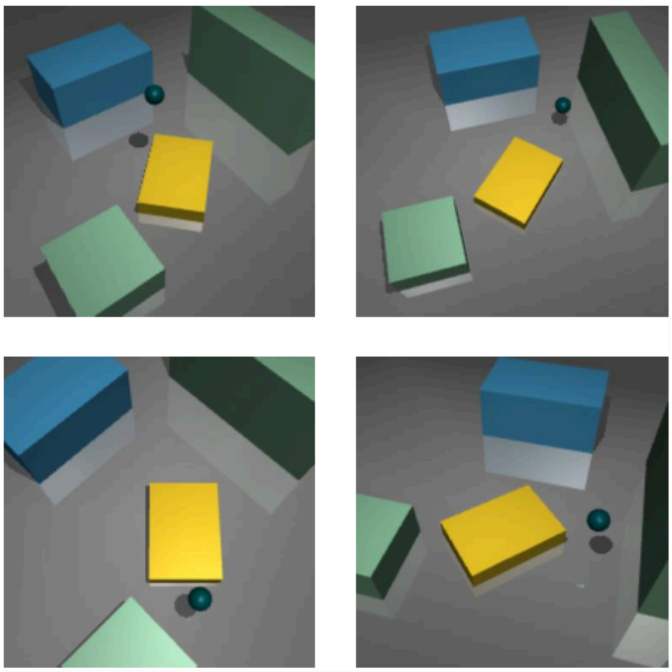
其中$V_W(\vec{\mathbf{r}}^{(W)})$描述从墙壁弹起，$V_O(\vec{\mathbf{r}}^{(W)})$描述从障碍物弹起。

After defining the potential for a specific physical situation, we implement the differential equation 5 in the PAF.

定义特定物理情况的势能后，我们在 PAF 中实现微分方程 5。

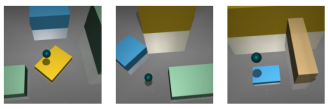
In our experiments, we use the 5 th order Dormand-Prince method [8] to solve the differential equations numerically, but we note that other solvers can be used as well.

在我们的实验中，我们使用五阶 Dormand-Prince 方法 [8] 来数值求解微分方程，但我们注意到也可以使用其他求解器。



(a) SD : images captured from different camera locations in the 1 st environment.

(a) SD：从第一环境中的不同相机位置捕获的图像。



(b) SD : images captured in different environments.

(b) SD：在不同环境中捕获的图像。



(c) RD : images captured from different camera locations.

(c) RD：从不同相机位置捕获的图像。

Figure 4 : Example images from the synthetic dataset (SD) and real dataset (RD).

图 4：来自合成数据集 (SD) 和真实数据集 (RD) 的示例图像。

3.5 Loss

3.5 损失

Since we do not use 3D ground truth labels, we utilize the heatmap loss and the future loss to train the PEN.

由于我们不使用 3D 真值标签，因此我们利用热图损失和未来预测损失来训练 PEN。

As it is common in human pose estimation (e.g. [17]), we also use heatmaps to teach the model to predict the 2D image coordinates.

由于这在人体姿态估计中很常见（例如[17]），我们还使用热图来教导模型预测 2D 图像坐标。

We calculate the L_2 loss between the heatmap output $\tilde{H}(t_n)$ of the PEN for the frame at time t_n and the ground truth heatmap $\tilde{H}_{\mathrm{gt}}(t_n)$.

我们计算t_n时刻帧的PEN热图输出$\tilde{H}(t_n)$与真值热图$\tilde{H}_{\mathrm{gt}}(t_n)$之间的$L_2$损失。

This ground truth heatmap is a two-dimensional Gaussian centered at the ground truth image coordinates at time t_n .

该真值热图是以t_n时刻的真值图像坐标为中心的二维高斯分布。

The future loss utilizes the prediction of the PAF to teach the model the correct prediction of the camera depth.

未来预测损失利用 PAF 的预测来教导模型正确预测相机深度。

First, we transform the prediction of the PAF at time $t_{n+\Delta n}$ to image coordinates using the camera matrices and obtain the image coordinates $\vec{\mathbf{r}}^{(I)}_{\mathrm{PAF}}$.

首先，我们使用相机矩阵将$t_{n+\Delta n}$时刻PAF的预测转换为图像坐标，并获得图像坐标$\vec{\mathbf{r}}^{(I)}_{\mathrm{PAF}}$。

Then, we calculate the L_1 loss between these coordinates and the image coordinates calculated by the PEN for the frame at time $t_{n+\Delta n}$.

然后，我们计算这些坐标与 PEN 在$t_{n+\Delta n}$时间帧计算的图像坐标之间的L_1损失。

Consequently, the total loss is calculated as

因此，总损失计算为

$$\begin{aligned} L &= \|\vec{\mathbf{r}}^{(I)}_{\mathrm{PEN}}(t_{n+\Delta n}) - \vec{\mathbf{r}}^{(I)}_{\mathrm{PAF}}(t_{n+\Delta n})\|_{\mathrm{L1}} \\ &+ \frac{1}{|\mathcal{T}|}\sum_{t_i \in \mathcal{T}} \|\tilde{H}(t_i) - \tilde{H}_{\mathrm{gt}}(t_i)\|_{\mathrm{L2}} \end{aligned} \tag{9}$$

with $\mathcal{T} = \{t_{n-1}, t_n, t_{n+1}, t_{n+\Delta n}\}$.

与$\mathcal{T} = \{t_{n-1}, t_n, t_{n+1}, t_{n+\Delta n}\}$。

3.6 Evaluation Metrics

3.6 评估指标

In order to test the quality of the PEN’s 3D predictions we introduce the distance to groundtruth (DtG) metric.

为评估 PEN 的三维预测质量，我们定义预测位置到真值的距离（distance to ground truth，DtG）。

It measures the euclidian distance between the prediction and the ground truth 3D position.

它测量预测和真值 3D 位置之间的欧氏距离。

Accordingly, the metric is calculated as

因此，该指标计算如下

$$\mathrm{DtG} = \frac{1}{|\mathcal{V}|}\sum_{t_i \in \mathcal{V}} \left\| \vec{\mathbf{r}}^{(C)}_{\mathrm{gt}}(t_i) - \vec{\mathbf{r}}^{(C)}_{\mathrm{PEN}}(t_i) \right\|_{\mathrm{L2}} \tag{10}$$

over all images in the test set $\mathcal{V}$.

测试集中的所有图像$\mathcal{V}$。

We note that the metric remains the same regardless of whether world coordinates or camera coordinates are used.

我们注意到，无论使用世界坐标还是相机坐标，度量都保持不变。

Since we expect the estimation of the ball’s 3D position to be less accurate if the ball is further away from the camera, we define the binned distance to groundtruth (DtG b) metric.

球距离相机越远，三维位置估计通常越不准确。因此，我们定义按深度分箱的真值距离指标（DtG b）。

Depending on the ball’s groundtruth camera depth $z^{(C)}_{\mathrm{gt}}$, the predictions are grouped into b bins and the DtG is calculated for each bin separately.

根据球的真实相机深度$z^{(C)}_{\mathrm{gt}}$，预测被分组为b个箱，并分别计算每个箱的 DtG。

Thus, it is calculated as

因此，计算为

$$\mathrm{DtG}_b = \begin{pmatrix} \frac{1}{|\mathcal{V}_0|} \sum_{t_i \in \mathcal{V}_0} \left\| \vec{\mathbf{r}}_{\mathrm{gt}}^{(C)}(t_i) - \vec{\mathbf{r}}_{\mathrm{PEN}}^{(C)}(t_i) \right\|_{\mathrm{L2}} \\ \vdots \\ \frac{1}{|\mathcal{V}_{b-1}|} \sum_{t_i \in \mathcal{V}_{b-1}} \left\| \vec{\mathbf{r}}_{\mathrm{gt}}^{(C)}(t_i) - \vec{\mathbf{r}}_{\mathrm{PEN}}^{(C)}(t_i) \right\|_{\mathrm{L2}} \end{pmatrix} \tag{11}$$

with $\mathcal{V}_j = \{t_i \in \mathcal{V} | z_{\mathrm{gt}}^{(C)}(t_i) \in [j \cdot \frac{z_{\max} - z_{\min}}{b} + z_{\min}, (j+1) \cdot \frac{z_{\max} - z_{\min}}{b} + z_{\min}]\}$ and $j \in \{0, ..., b-1\}$.

与$\mathcal{V}_j = \{t_i \in \mathcal{V} | z_{\mathrm{gt}}^{(C)}(t_i) \in [j \cdot \frac{z_{\max} - z_{\min}}{b} + z_{\min}, (j+1) \cdot \frac{z_{\max} - z_{\min}}{b} + z_{\min}]\}$和$j \in \{0, ..., b-1\}$。

We select suitable values for the number of bins b , the minimal distance $z_{\min}$ and the maximal distance $z_{\max}$ depending on each dataset.

我们根据每个数据集为 bin 数量b、最小距离$z_{\min}$和最大距离$z_{\max}$选择合适的值。

Table 1: DtG scores for the synthetic dataset .

表 1：合成数据集的 DtG 分数。

(a) DtG scores per camera location evaluated on the 1 st environment.

(a) 在第一个环境中评估的每个相机位置的 DtG 分数。

	DtG ± ΔDtG (cm) ↓				DtG ± ΔDtG (cm) ↓	
training set 训练集	camera 1 相机1	camera 7 相机7	camera 8 相机8	camera 9 相机9	**training set** 训练集	env 1 环境1	env 2 环境2
SD-S **SD-S**	22 ± 19	-	-	-	**SD-S** **SD-S**	-	-
SD-M **SD-M**	19 ± 10	27 ± 23	23 ± 9	21 ± 10	**SD-M** **SD-M**	22 ± 16	-
SD-L **SD-L**	11 ± 6	28 ± 25	15 ± 8	16 ± 7	**SD-L** **SD-L**	18 ± 16	17 ± 16

(b) DtG scores per environment.

(b) 每个环境的 DtG 分数。

4 Dataset

4 数据集

To test our method we create two datasets consisting of monocular videos of a moving ball: the synthetic dataset (SD) and the real dataset (RD) .

为了测试我们的方法，我们创建了两个由移动球的单目视频组成的数据集：合成数据集（SD）和真实数据集（RD）。

For the training we provide the ball’s 2D positions for every frame and for the evaluation we also provide the ball’s 3D position.

对于训练，我们提供每一帧的球的 2D 位置，对于评估，我们还提供球的 3D 位置。

Some example images are depicted in Figure 4 and more images are visualized in the supplementary material.

图 4 中描绘了一些示例图像，补充材料中显示了更多图像。

4.1 Synthetic Dataset

4.1 合成数据集

We create the SD using the general purpose physics engine MuJoCo [29] and, thus, we are able to create synthetic data with realistic physical behavior.

我们使用通用物理引擎 MuJoCo [29] 构建 SD，从而生成具有真实物理行为的合成数据。

We simulate a moving ball under the influence of gravity.

我们模拟在重力影响下移动的球。

The ball bounces off an infinite floor and additional finite obstacles placed in the environment.

球从无限的地面和环境中放置的其他有限障碍物上弹起。

We generate 3 different environments and for each environment we capture the scenes from 9 different camera location.

我们生成 3 个不同的环境，对于每个环境，我们从 9 个不同的摄像机位置捕获场景。

However, each distinct scene is only captured from a single camera in contrast to traditional triangulation and binocular vision settings.

然而，与传统的三角测量和双目视觉设置相比，每个不同的场景仅由单个摄像机捕获。

The cameras are placed such that the distance to the origin of the scene is between $8\,m$ and $10\,m$, and the exact camera locations are given in the supplementary material.

摄像机的放置使得到场景原点的距离在$8\,m$和$10\,m$之间，并且确切的摄像机位置在补充材料中给出。

The videos are recorded with $30\,\mathrm{FPS}$.

视频以$30\,\mathrm{FPS}$录制。

We create 3 versions of the dataset:

我们创建了 3 个版本的数据集：

- SD-S : We use a single environment and capture the scene from a single fixed camera location.

  SD-S：我们使用单一环境并从单一固定摄像机位置捕获场景。

  We create 100 videos for the training set, 100 for the validation set and 100 for the test set.

  我们为训练集创建 100 个视频，为验证集创建 100 个视频，为测试集创建 100 个视频。

- SD-M : We use a single environment and capture the scene from 9 different camera locations.

  SD-M：我们使用单一环境并从 9 个不同的摄像机位置捕获场景。

  The training set consists of 100 videos per camera location recorded from camera locations 1-6, while the validation and test set each contain 100 videos per camera location recorded from all 9 camera locations.

  训练集包含从摄像机位置 1-6 记录的每个摄像机位置 100 个视频，而验证集和测试集分别包含从所有 9 个摄像机位置记录的每个摄像机位置 100 个视频。

- SD-L : We include videos recorded in 3 different environments and for each environment the scene can be captured from 9 different camera locations.

  SD-L：我们包括在 3 个不同环境中录制的视频，对于每个环境，可以从 9 个不同的摄像机位置捕获场景。

  For each environment, the training set contains 100 videos per camera location recorded from camera locations 1-6, while the validation and test set contain 100 videos per camera location recorded from all 9 camera locations.

对于每个环境，训练集包含从摄像机位置 1-6 记录的每个摄像机位置 100 个视频，而验证和测试集包含从所有 9 个摄像机位置记录的每个摄像机位置 100 个视频。

The versions are designed such that SD-M is a subset of SD-L and SD-S is a subset of SD-M .

这些版本的设计使得 SD-M 是 SD-L 的子集，SD-S 是 SD-M 的子集。

Each image has a resolution of 224×224 pixel and each clip consists of 30 consecutive frames during training.

每个图像的分辨率为224×224像素，训练期间每个剪辑由 30 个连续帧组成。

Example frames for different camera locations in the 1 st environment are shown in Figure 4(a) , and for the three different environments in Figure 4(b) .

图 4(a) 中显示了第一环境中不同相机位置的示例帧，图 4(b) 中显示了三个不同环境中的示例帧。

Random initial positions and velocities of the ball are selected for each video to ensure diversity among the samples.

为每个视频选择随机的球初始位置和速度，以确保样本之间的多样性。

By applying our method to SD-S we are able to show that our model is capable of learning the ball’ s 3D position in general.

在 SD-S 上应用该方法的实验表明，模型具备学习球的三维位置的基本能力。

However, real use cases like sport broadcasting might not use a fixed camera, instead the camera sometimes moves and changes its location between the scenes.

然而，像体育转播这样的实际用例可能不会使用固定摄像机，而是摄像机有时会在场景之间移动并改变其位置。

Therefore, we use the SD-M to show that our model is also able to learn the ball’ s 3D position if the camera changes its location.

因此，我们使用 SD-M 来表明，如果相机位置发生变化，我们的模型也能够学习球的 3D 位置。

Finally, it is beneficial for practical applications to train a model that is working on different environments like different known courts in sport broadcasting.

最后，训练一个在不同环境（例如体育转播中不同的已知球场）上工作的模型对于实际应用是有益的。

Using the SD-L , we show that our model is also able to learn the ball’ s 3D position even if the environment changes.

使用 SD-L，我们表明即使环境发生变化，我们的模型也能够学习球的 3D 位置。

By applying our method to these three synthetic datasets we test some characteristics needed for practical application.

通过将我们的方法应用于这三个合成数据集，我们测试了实际应用所需的一些特征。

4.2 Real Dataset

4.2 真实数据集

Since the synthetic images are generated by a physics engine, they are visually not perfectly realistic.

由于合成图像是由物理引擎生成的，因此它们在视觉上并不完全真实。

Therefore, we create the RD to prove that our method is also able to cope with noisy input data.

因此，我们创建 RD 来证明我们的方法也能够处理含噪声的输入数据。

We use a ZED 2i stereo camera [28] to record a rubber ball bouncing off the floor, the walls and an additional obstacle.

我们使用 ZED 2i 立体摄像机 [28] 记录橡胶球从地面、墙壁和其他障碍物弹起的情况。

We record each video from one of two distinct camera locations and use a frame rate of $60\ \mathrm{FPS}$.

我们从两个不同的摄像机位置之一记录每个视频，并使用$60\ \mathrm{FPS}$的帧速率。

The distance of the camera to the origin of the scene is between $1.3\ m$ and $1.4\ m$.

摄像机到场景原点的距离在$1.3\ m$和$1.4\ m$之间。

Only the data of the left camera is used as input to the PEN, and we manually labelled the 2D image coordinates for each frame.

仅使用左摄像头的数据作为 PEN 的输入，并且我们手动标记每一帧的 2D 图像坐标。

We depict example frames from the two different camera locations in Figure 4(c) .

我们在图 4(c) 中描绘了来自两个不同相机位置的示例帧。

Each frame is resized to 224×384 pixel.

每帧的大小都调整为224×384像素。

We split the videos into clips consisting of 16 frames each, with 251 clips in the train set, 34 clips in the validation set, and 81 clips in the test set.

我们将视频分成每个由 16 帧组成的剪辑，训练集中有 251 个剪辑，验证集中有 34 个剪辑，测试集中有 81 个剪辑。

In order to evaluate our method we use the data of the stereo camera to calculate a ground truth depthmap of the scene.

为了评估我们的方法，我们使用立体相机的数据来计算场景的真值深度图。

The ground truth camera depth is obtained from the depthmap value at the 2D image coordinates of the ball.

相机深度的真值是从球的 2D 图像坐标处的深度图值获得的。

By applying our method to this dataset we are able to show that it also works with simple yet realistic input data.

在该数据集上的实验表明，本方法同样适用于简单但贴近真实情况的输入数据。

5 Experiments

5 实验

In this section we perform multiple experiments and apply our method to the datasets described in section 4 .

在本节中，我们执行多个实验并将我们的方法应用于第 4 节中描述的数据集。

We use the first 4 stages of a ResNet34 [14] pretrained on ImageNet [27] as the backbone of the PEN and provide additional experiments with different backbones in the supplementary material.

我们使用在 ImageNet [27] 上预训练的 ResNet34 [14] 的前 4 个阶段作为 PEN 的主干，并在补充材料中提供不同主干的额外实验。

To ensure performance stability, we maintain a second copy of our model whose weights are an exponential moving average (EMA) of the trained model weights, and we update the EMA model after each iteration.

为了确保性能稳定性，我们维护模型的第二个副本，其权重是训练模型权重的指数移动平均值 (EMA)，并且我们在每次迭代后更新 EMA 模型。

At the end of each epoch the EMA model is evaluated on the validation set, and we only keep the model with the best DtG score.

在每个 epoch 结束时，EMA 模型都会在验证集上进行评估，并且我们只保留具有最佳 DtG 分数的模型。

We calculate the future loss at multiple time steps.

我们计算多个时间步长的未来预测损失。

For the synthetic data we use $\Delta n = 4$, $\Delta n = 8$ and $\Delta n = 15$ and for the real data we use $\Delta n = 2$, $\Delta n = 4$ and $\Delta n = 6$, because the average speed of the real ball is larger than the speed of the synthetic ball.

对于合成数据，我们使用$\Delta n = 4$，$\Delta n = 8$和$\Delta n = 15$，对于真实数据，我们使用$\Delta n = 2$，$\Delta n = 4$和$\Delta n = 6$，因为真实球的平均速度大于合成球的速度。

Our code and datasets will be published at www.example.com where we will also provide additional visual evaluations of our experiments.

我们的代码和数据集将发布在 www.example.com 上，我们还将在那里提供对我们的实验的额外视觉评估。

5.1 Synthetic Experiments

5.1 合成实验

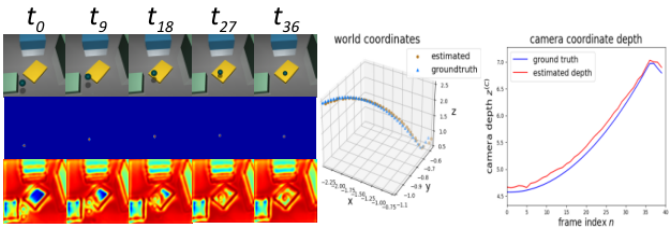


Figure 5 : Predictions of the SD -L model on a sample video from the 1st camera location in the 1st environment.

图 5：SD-L 模型对来自1st环境中1st摄像机位置的示例视频的预测。

The first row shows the input images, the second row shows the predicted heatmaps, and the third row shows the predicted depthmaps.

第一行显示输入图像，第二行显示预测的热图，第三行显示预测的深度图。

On the right, a 3D plot of the predicted ball trajectory $\vec{\mathbf{r}}^{(W)}$ and a 2D plot of the camera depth $z^{(C)}$ is shown.

右侧显示了预测球轨迹的 3D 图$\vec{\mathbf{r}}^{(W)}$和相机深度的 2D 图$z^{(C)}$。

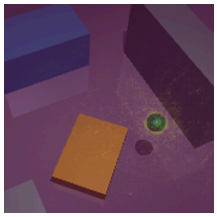


Figure 6 : Saliency Map for a prediction of the SD-L model.

图 6：SD-L 模型预测的显著图。

In this section we evaluate our method on the synthetic datasets described in section 4 .

在本节中，我们在第 4 节中描述的合成数据集上评估我们的方法。

We train three models, one for each subset of the synthetic datasets:

我们训练三个模型，每个模型对应合成数据集的每个子集：

- SD-S : The first model is trained solely on the SD-S training set.

  SD-S：第一个模型仅在 SD-S 训练集上进行训练。

  Therefore, the model is only exposed to images from a single camera location and a single environment during both training and testing.

  因此，在训练和测试期间，模型仅暴露于来自单个摄像机位置和单个环境的图像。

- SD-M : The second model is trained on the SD-M training set, which includes images from the first 6 camera locations of the 1 st environment.

  SD-M：第二个模型在 SD-M 训练集上进行训练，其中包括来自第一个环境的前 6 个摄像机位置的图像。

  However, during evaluation, it is tested on videos recorded from all camera locations of the 1 st environment.

  然而，在评估过程中，它是在从第一个环境的所有摄像机位置录制的视频上进行测试的。

  By comparing its results on camera 1 to the results of the first model, we test if training on multiple camera locations improves the model’ s performance.

  通过将相机 1 上的结果与第一个模型的结果进行比较，我们测试在多个相机位置上的训练是否可以提高模型的性能。

By looking at the results on the last three camera locations we test if the model is able to generalize to previously unseen camera locations.

通过查看最后三个摄像机位置的结果，我们测试模型是否能够泛化到以前未见过的摄像机位置。

-

SD-L : The third model is trained on the SD-L training set, which includes images from the first 6 camera locations of each environment.

SD-L：第三个模型在 SD-L 训练集上进行训练，其中包括来自每个环境的前 6 个摄像机位置的图像。

During evaluation, it is tested on all camera locations of each environment in the test set.

在评估过程中，会在测试集中每个环境的所有相机位置上进行测试。

With this model we are able to test whether the model is able to learn the ball’ s position for different known environments.

通过这个模型，我们可以测试该模型是否能够学习不同已知环境下球的位置。

Moreover, we test if the model benefits from being trained on multiple environments by comparing the results of this model to the results of the second model on the 1 st environment.

此外，我们通过将该模型的结果与第二个模型在第一个环境上的结果进行比较来测试模型是否受益于在多个环境中进行训练。

The results on the test set are given in Table 1 .

测试集的结果如表1所示。

For the three models, we evaluate the average DtG sores on the videos from exemplary camera locations in the 1 st environment in Table 1(a) .

表 1(a) 比较了这 3 个模型在第 1 个环境中若干代表性相机位置上的平均 DtG。

Since the performance of the SD-M model is significantly better than the performance of the SD-S model, we conclude that the PEN is not just able to deal with input from multiple camera perspectives, but it even benefits from being trained with additional data from multiple camera location.

由于 SD-M 模型的性能明显优于 SD-S 模型的性能，因此我们得出结论，PEN 不仅能够处理来自多个摄像机角度的输入，而且甚至受益于使用来自多个摄像机位置的附加数据进行训练。

Moreover, we see that the model is able to generalize to previously unseen camera locations since the scores for camera 7 to 9 are also good.

此外，我们看到该模型能够泛化到以前未见过的相机位置，因为相机 7 到 9 的分数也很好。

By comparing the SD-L model to the SD-M model, we see that the SD-L model achieves even slightly better scores and, consequently, we conclude that it is possible to apply the PEN to multiple known environments.

通过将 SD-L 模型与 SD-M 模型进行比较，我们发现 SD-L 模型取得了稍微更好的分数，因此我们得出结论，可以将 PEN 应用于多个已知环境。

In Table 1(b) we see the average DtG scores on the three environments.

在表 1(b) 中，我们看到了三种环境的平均 DtG 分数。

The SD-L model performs again better than the SD-M model, thus, reinforcing the impression that training the model on multiple environments is beneficial.

SD-L 模型的表现再次优于 SD-M 模型，因此强化了在多种环境下训练模型是有益的这一印象。

The results of the SD-L model are visualized in Figure 5 , and it is visible that the predictions match the 3D ground truth very accurately.

SD-L 模型的结果如图 5 所示，可以看出预测与 3D 真值非常准确地匹配。

To conclude, we see that it is not only possible to train our model with distinct videos from multiple camera locations, but even recognize the fact that using distinct videos from multiple camera locations is beneficial.

总而言之，我们发现不仅可以使用来自多个摄像机位置的不同视频来训练我们的模型，而且甚至认识到使用来自多个摄像机位置的不同视频是有益的。

This is important for the application to realistic data (e.g. sports broadcasting) as in many cases the camera is changing its position instead of being stationary.

这对于现实数据（例如体育转播）的应用非常重要，因为在许多情况下摄像机正在改变其位置而不是静止的。

Moreover, we also show that our method is capable of dealing with multiple known environments, which is also an important property for practical applications (e.g. different stadiums or courts in sport broadcasting).

此外，我们还表明我们的方法能够处理多种已知环境，这也是实际应用的一个重要特性（例如体育转播中的不同体育场或球场）。

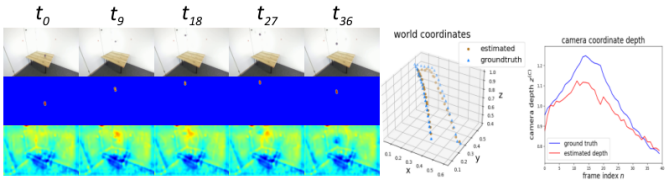


Figure 7 : PEN’ s predictions on a sequence from the real dataset .

图 7：PEN 对真实数据集序列的预测。

The first row shows the input images, the second row shows the predicted heatmaps, and the third row shows the predicted depth map.

第一行显示输入图像，第二行显示预测的热图，第三行显示预测的深度图。

On the right, a 3D plot of the predicted ball trajectory $\vec{\mathbf{r}}^{(W)}$ and the camera depth $z^{(C)}$ is shown.

右侧显示了预测球轨迹$\vec{\mathbf{r}}^{(W)}$和相机深度$z^{(C)}$的 3D 图。

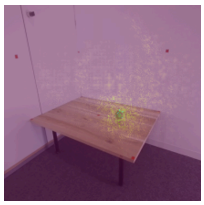


Figure 8 : Saliency map for a prediction of the RD model.

图 8：RD 模型预测的显著图。

5.2 Real Experiments

5.2 真实实验

Even though we were able to experimentally prove some properties needed for practical applications on the synthetic dataset , we still need to show that our model is able to work with real data.

尽管合成数据集上的实验已验证了实际应用所需的一些性质，我们仍需检验模型能否处理真实数据。

Important difficulties in working with real videos are that the model has to be able to filter out pixel noise in the image data and that it needs to be capable of working with noisy labels.

处理真实视频的重要困难在于，模型必须能够滤除图像数据中的像素噪声，并且需要能够处理噪声标签。

Because the 2D image labels are manually annotated and do not always perfectly match the center of the ball, additional noise is also added to the calculation of the future loss.

由于 2D 图像标签是手动标注的，并不总是与球的中心完美匹配，因此在未来预测损失的计算中也会添加额外的噪声。

Furthermore, the physical description of the ball might not perfectly match the ball’ s real behavior (e.g. due to deformations of the ball or spin), resulting in reduced precision of the forecasting.

此外，球的物理描述可能与球的真实行为不完全匹配（例如，由于球的变形或旋转），导致预测的精度降低。

To experimentally demonstrate that our model is able to cope with these difficulties, we evaluate our method on the real dataset .

为了通过实验证明我们的模型能够应对这些困难，我们在真实数据集上评估了我们的方法。

Table 2: Scores per camera location on the real dataset .

表 2：真实数据集上每个相机位置的得分。

	$\mathrm{DtG} \pm \Delta\mathrm{DtG}$ (cm) ↓		$\mathrm{DtG}_3 \pm \Delta\mathrm{DtG}_3$ (cm) ↓	
training set	camera 1	camera 2	camera 1	camera 2
训练集	相机1	相机2	相机1	相机2
RD	7 ± 4	6 ± 4	$\begin{pmatrix} 8 \pm 4 \\ 6 \pm 2 \\ 11 \pm 4 \end{pmatrix}$	$\begin{pmatrix} 4 \pm 2 \\ 6 \pm 3 \\ 11 \pm 4 \end{pmatrix}$

The results on the test set of the real dataset are shown in Table 2 .

真实数据集测试集上的结果如表2所示。

We achieve a DtG score of $7\ \mathrm{cm}$ for the images recorded from the 1 st camera location and a DtG score of $6\ \mathrm{cm}$ for the images from the 2 nd camera location, thus achieving very precise predictions.

我们对第一个摄像机位置记录的图像实现了$7\ \mathrm{cm}$的 DtG 分数，对第二个摄像机位置记录的图像实现了$6\ \mathrm{cm}$的 DtG 分数，从而实现了非常精确的预测。

Consequently, our method is able to deal with the difficulties of real data.

因此，我们的方法能够处理真实数据的困难。

Furthermore, we calculate the DtG_3 scores by dividing the data into 3 bins according to equation 11 with $z_{\min} = 5\ \mathrm{cm}$ and $z_{\max} = 2\ m$.

此外，我们根据公式 11、$z_{\min} = 5\ \mathrm{cm}$和$z_{\max} = 2\ m$将数据分为 3 个区间，计算出$\mathrm{DtG}_3$分数。

By analyzing the individual bins, we see that the scores in the 3 rd bin are noticeably worse.

通过分析各个箱，我们发现第三个箱的分数明显更差。

This can be explained by the fact that it is harder for the model to predict the ball’ s location when it is farther away from the camera.

这可以用以下事实来解释：当球距离相机较远时，模型很难预测球的位置。

However, the predictions are still very accurate in the third bin.

然而，第三个 bin 的预测仍然非常准确。

Nevertheless, further research on this scaling behavior is needed for the next step towards future applications.

尽管如此，为了未来的应用，还需要对这种缩放行为进行进一步的研究。

In figure 7 we show the predictions of the model for an example video sequence.

在图 7 中，我们显示了模型对示例视频序列的预测。

Looking at the 3D plot, it is also clear that the predicted trajectory of the ball matches the ground truth trajectory very closely.

查看 3D 图，还可以清楚地看到球的预测轨迹与真值轨迹非常接近。

5.3 Emergence of Depthmaps

5.3 深度图的出现

Figure 5 and 7 clearly reveal the emergence of the scene’ s geometric structure in the depthmaps, despite the fact that precise depth prediction is only necessary in the vicinity of the ball to produce accurate outputs.

图 5 和图 7 清楚地揭示了深度图中场景几何结构的出现，尽管事实上只有在球附近才需要精确的深度预测才能产生准确的输出。

Thus, it appears that the model is able to learn a full depthmap of the environment.

因此，该模型似乎能够学习环境的完整深度图。

This observation highlights an intriguing characteristic of our method that raises the possibility of indirect training for monocular depth estimation.

这一观察结果凸显了我们的方法的一个有趣的特征，它提高了单目深度估计间接训练的可能性。

However, a comprehensive analysis of this phenomenon is beyond the scope of this paper, and future research is necessary to evaluate this aspect in more detail.

然而，对这一现象的全面分析超出了本文的范围，未来的研究需要更详细地评估这方面。

5.4 Interpretation of the learning process

5.4 学习过程解读

In this section, we delve into the question of what information the PEN utilizes in the images to calculate the depth.

在本节中，我们将深入研究 PEN 利用图像中的哪些信息来计算深度的问题。

It is possible that the network simply measures the diameter of the ball in the image for the calculation of the depth (similar to [30]).

网络可能只是简单地测量图像中球的直径来计算深度（类似于[30]）。

Another possibility is that the network compares the ball to its surrounding environment.

另一种可能性是网络将球与其周围环境进行比较。

To gain a rough understanding of the regions in the image that are important for the calculation of the depth, we calculate a saliency map S for the input images as

为了粗略地了解图像中对深度计算很重要的区域，我们为输入图像计算显著图S：

$$S_{i,j} = \sum_{c=0}^{3} \frac{\partial z^{(C)}}{\partial I_{c,i,j}} \tag{12}$$

where I is the input image, $z^{(C)}$ is the predicted depth of the ball, i and j index the pixels of the image, and c denotes the channel index.

其中I是输入图像，$z^{(C)}$是球的预测深度，i和j是图像的像素索引，c表示通道索引。

In Figure 6 and 8 we depict the saliency map for a real as well as a synthetic image.

在图 6 和图 8 中，我们描述了真实图像和合成图像的显著图。

We observe the highest values around the ball, indicating that the PEN focuses mainly on the size of the ball for the estimation of the depth.

我们观察到球周围的最高值，表明 PEN 主要关注球的大小来估计深度。

However, we also see that the PEN considers the local surroundings of the ball, suggesting that this additional information enhances the depth estimation beyond a simple diameter measurement.

然而，我们还看到 PEN 考虑了球的局部环境，这表明这些附加信息增强了深度估计，而不仅仅是简单的直径测量。

Furthermore, we conjecture that the learned depthmap of the scene is a result of the PEN attending to the local context around the ball.

此外，我们推测学习到的场景深度图是 PEN 关注球周围局部上下文的结果。

6 Conclusion

6 结论

This paper has introduced a novel approach for monocular 3D object localization and demonstrated its effectiveness through experiments on both synthetic and real datasets.

本文介绍了一种新的单目 三维目标定位方法，并通过对合成数据集和真实数据集的实验证明了其有效性。

Our proposed method eliminates the need for 3D labels as supervision by leveraging the physical equations of motion, allowing the model to infer the latent third dimension.

我们提出的方法通过利用运动物理方程消除了 3D 标签作为监督的需要，允许模型推断潜在的三维。

We discussed various properties relevant to real-world scenarios and demonstrated the robustness of our method in handling them.

我们讨论了与现实场景相关的各种属性，并证明了我们的方法在处理它们时的稳健性。

Consequently, our method is a promising step towards 3D object localization applications, although further research is needed to enhance its practical usability.

因此，我们的方法是迈向 三维目标定位应用的有希望的一步，尽管需要进一步的研究来增强其实际可用性。

Therefore, we intend to conduct further research to refine and expand upon our method, focusing on areas like scalability, interpretability, and generalization ability.

因此，我们打算进行进一步的研究来完善和扩展我们的方法，重点关注可扩展性、可解释性和泛化能力等领域。

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Supplementary Material

补充材料

In this supplementary material, we provide additional details on the methodology, offer further descriptions and visualizations of the datasets, and present additional results.

在本补充材料中，我们提供了有关该方法的更多详细信息，提供了数据集的进一步描述和可视化，并提供了其他结果。

The code and dataset is published at https://kiedani.github.io/3DV2024/ .

代码和数据集发布在 https://kiedani.github.io/3DV2024/ 。

7 Methods

7 方法

7.1 Derivation of the Differential Equations of Motion

7.1 微分运动方程的推导

In order to define the differential equations of motion 5 , we first need to model the full potential 8 .

为了定义运动微分方程 5 ，我们首先需要对全部势能 8 进行建模。

While the gravitational potential is already defined in equation 6 and the bouncing off the floor is modeled through equation 7 , we still need to define V_W and V_O .

虽然重力势已在方程 6 中定义，并且通过方程 7 对地面弹跳进行了建模，但我们仍然需要定义V_W和V_O。

The potential V_W models the interaction of the ball with infinite walls of the room and, thus, it is very similar to V_f .

势V_W模拟了球与房间无限墙壁的相互作用，因此，它与V_f非常相似。

Since there are no walls in the synthetic dataset, we set $V_W = 0$.

由于合成数据集中没有墙，我们设置$V_W = 0$。

In the real dataset are two walls, one in the y-z plane at $x = 0\ m$ and one in the x-z plane at $y = 0.6\ m$.

真实数据集中有两堵墙，一堵墙位于y-z平面的$x = 0\ m$处，一堵墙位于x-z平面的$y = 0.6\ m$处。

Thus, the ball is constrained by the walls and the floor to the area $x \geq 0\ m$, $y \leq 0.6\ m$ and $z \geq 0\ m$.

因此，球被墙壁和地面限制在$x \geq 0\ m$、$y \leq 0.6\ m$和$z \geq 0\ m$区域。

Therefore, we model the potential V_W as

因此，我们将势V_W建模为

$$V_W(\vec{\mathbf{r}}^{(W)}) = mc \cdot \mathrm{ReLU}(-x^{(W)}) + mc \cdot \mathrm{ReLU}(y^{(W)} - 0.6\ m) \tag{13}$$

with $c = 1000$ being a large constant.

其中$c = 1000$是一个大常数。

The collisions with finite obstacles are described by the potential V_O , and there are multiple obstacles in the three different environments of the synthetic dataset and one obstacle in the real dataset.

与有限障碍物的碰撞由势V_O来描述，并且在合成数据集的三个不同环境中存在多个障碍物，而在真实数据集中存在一个障碍物。

To model the potential, we consider a cubic object lying on the floor at the origin of the so-called object coordinate system with the length $2l$, width $2w$ and height h such that the object coordinates $\vec{\mathbf{r}}^{(o)} = \begin{pmatrix} x^{(o)} & y^{(o)} & z^{(o)} \end{pmatrix}^T$ with $x^{(o)} \in [-l, l] \wedge y^{(o)} \in [-w, w] \wedge z^{(o)} \in [0, h]$ describe points inside the cube.

为构建势能模型，考虑一个放在地面上的长方体，其物体坐标系原点位于长方体中心，长、宽、高分别为 $2l$、$2w$、h；满足 $x^{(o)} \in [-l, l] \wedge y^{(o)} \in [-w, w] \wedge z^{(o)} \in [0, h]$ 的物体坐标 $\vec{\mathbf{r}}^{(o)} = \begin{pmatrix} x^{(o)} & y^{(o)} & z^{(o)} \end{pmatrix}^T$ 对应长方体内部的点。

To accurately model collisions with this object, the potential has to be large around the boundaries of the cube and zero everywhere outside the cube.

为了准确地模拟与该对象的碰撞，立方体边界周围的势必须很大，而立方体外部的任何地方都必须为零。

In this situation, the potential V_o describing collisions with this single obstacle at the origin can be described as

此时，原点处单个障碍物对应的碰撞势能 V_o 可写为

$$\begin{aligned} V_o(\vec{\mathbf{r}}^{(o)}) \quad = m\frac{c}{\beta_o} \cdot & \left(\sigma\left(\beta_o(x^{(o)} + l)\right) - \sigma\left(\beta_o(x^{(o)} - l)\right)\right) \\ \cdot & \left(\sigma\left(\beta_o(y^{(o)} + w)\right) - \sigma\left(\beta_o \cdot (y^{(o)} - w)\right)\right) \\ \cdot & \left(\sigma\left(\beta_o \cdot (z^{(o)})\right) - \sigma\left(\beta_o \cdot (z^{(o)} - h)\right)\right) \end{aligned} \tag{14}$$

using the sigmoid function $\sigma(\cdot)$ and the large constant $\beta_o = 66$ to sharpen the softmax function.

使用sigmoid函数$\sigma(\cdot)$和大常数$\beta_o = 66$来锐化softmax函数。

In order to describe multiple obstacles at different known positions, we have to transform the ball’ s world coordinates $\vec{\mathbf{r}}^{(W)}$ into the object coordinates $\vec{\mathbf{r}}^{(o)}$ for each obstacle separately such that the obstacle is centered at the origin of the object coordinate system.

为了描述不同已知位置的多个障碍物，我们必须将球的世界坐标$\vec{\mathbf{r}}^{(W)}$分别转换为每个障碍物的物体坐标$\vec{\mathbf{r}}^{(o)}$，使得障碍物以物体坐标系原点为中心。

This transformation can be realized using rotation and translation and is described by the transformation $\vec{\mathbf{r}}^{(o)} = \mathcal{T}_o(\vec{\mathbf{r}}^{(W)})$ for each obstacle o .

这种变换可以使用旋转和平移来实现，并且通过每个障碍物o的变换$\vec{\mathbf{r}}^{(o)} = \mathcal{T}_o(\vec{\mathbf{r}}^{(W)})$来描述。

As a result, the potential V_O describing collisions with all obstacles in the specific environment is described as

因此，特定环境中所有障碍物共同对应的碰撞势能 V_O 为

$$V_O(\vec{\mathbf{r}}^{(W)}) = \sum_{o \in \mathcal{O}} V_o\left(\mathcal{T}_o(\vec{\mathbf{r}}^{(W)})\right) \tag{15}$$

with $\mathcal{O}$ being the set of all obstacles placed in the environment.

$\mathcal{O}$是环境中所有障碍物的集合。

With the potentials V_G , V_F , V_W and V_O defined, the differential equations of motion 5 can now be derived.

定义了势V_G、V_F、V_W和V_O，现在可以导出运动微分方程5。

This can be achieved using automatic differentiation, but we decided to calculate the derivatives of the Hamilton function analytically for the sake of a small speed up.

这可以通过自动微分完成；为略微加快计算，我们选择解析计算哈密顿函数的导数。

We approximate the derivative of the ReLU function with a sigmoid function as $\frac{\partial}{\partial x}\mathrm{ReLU}(x) \approx \sigma(\beta_{\mathrm{g/w/f}} \cdot x)$ with $\beta_{\mathrm{g/w/f}} = 200$ being a large constant.

我们用 sigmoid 函数近似 ReLU 函数的导数，即 $\frac{\partial}{\partial x}\mathrm{ReLU}(x) \approx \sigma(\beta_{\mathrm{g/w/f}} \cdot x)$，其中 $\beta_{\mathrm{g/w/f}} = 200$ 为较大的常数。

There are no walls in the synthetic dataset and, thus, we obtain the differential equations of motion as

合成数据集中没有墙，因此，我们获得运动微分方程为

$$\begin{aligned}
\frac{d}{\mathrm{dt}}\vec{\mathbf{r}}^{(W)} &= \vec{\mathbf{v}}\,, \\
\frac{d}{\mathrm{dt}}\vec{\mathbf{v}} &= -\begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}\sigma\left(\beta_{g/w/f}\vec{\mathbf{r}}^{(W)}\right) + \begin{pmatrix} 0 \\ 0 \\ c \end{pmatrix}\sigma\left(-\beta_{g/w/f}\vec{\mathbf{r}}^{(W)}\right) \\
&\quad + \sum_{o \in \mathcal{O}} \beta_o V_o(\vec{\mathbf{r}}^{(o)})\mathcal{T}_o \\
&\quad \cdot \left(1 - \sigma\left(\beta_o\left(\vec{\mathbf{r}}^{(o)} + \begin{pmatrix} l \\ w \\ 0 \end{pmatrix}\right)\right) - \sigma\left(\beta_o\left(\vec{\mathbf{r}}^{(o)} - \begin{pmatrix} l \\ w \\ h \end{pmatrix}\right)\right)\right).
\end{aligned} \tag{16}$$

For the real dataset we additionally consider the two walls and obtain the equations of motion as

对于真实数据集，我们另外考虑两个墙壁并获得运动方程为

$$\begin{aligned}
\frac{d}{\mathrm{dt}}\vec{\mathbf{r}}^{(W)} &= \vec{\mathbf{v}}\,, \\
\frac{d}{\mathrm{dt}}\vec{\mathbf{v}} &= -\begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}\sigma\left(\beta_{g/w/f}\vec{\mathbf{r}}^{(W)}\right) + \begin{pmatrix} 0 \\ 0 \\ c \end{pmatrix}\sigma\left(-\beta_{g/w/f}\vec{\mathbf{r}}^{(W)}\right) \\
&\quad + \begin{pmatrix} c \\ 0 \\ 0 \end{pmatrix}\sigma\left(-\beta_{g/w/f}\vec{\mathbf{r}}^{(W)}\right) + \begin{pmatrix} 0 \\ c \\ 0 \end{pmatrix}\sigma\left(\beta_{g/w/f}\left(\vec{\mathbf{r}}^{(W)} - 0.6\,m\right)\right) \\
&\quad + \sum_{o \in \mathcal{O}} \beta_o V_o(\vec{\mathbf{r}}^{(o)})\mathcal{T}_o \\
&\quad \cdot \left(1 - \sigma\left(\beta_o\left(\vec{\mathbf{r}}^{(o)} + \begin{pmatrix} l \\ w \\ 0 \end{pmatrix}\right)\right) - \sigma\left(\beta_o\left(\vec{\mathbf{r}}^{(o)} - \begin{pmatrix} l \\ w \\ h \end{pmatrix}\right)\right)\right).
\end{aligned} \tag{17}$$

In this paper we model the equations of motion directly.

本文直接对运动方程建模。

However, it is also possible to use a differential physics engine like BRAX [10] in the PAF.

也可在 PAF 中使用 BRAX [10] 等可微物理引擎。

This would also allow to describe some physical phenomenons like spin even more easily.

这也使得可以更容易地描述一些物理现象，例如旋转。

7.2 Obtaining the Camera Matrices

7.2 获取相机矩阵

We need the intrinsic as well as extrinsic camera matrix for our method.

本方法需要相机内参矩阵和外参矩阵。

The intrinsic camera matrix is defined as

相机内参矩阵定义为

$$\mathbf{M}_{int} = \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix} \tag{18}$$

with $f_{x/y}$ being the focal length and $c_{x/y}$ being the principal point.

其中$f_{x/y}$为焦距，$c_{x/y}$为主点。

With this matrix the euclidean camera coordinates can be converted into homogeneous image coordinates .

通过这个矩阵，欧几里得相机坐标可以转换为齐次图像坐标。

The extrinsic camera matrix is defined as

相机外参矩阵定义为

$$\mathbf{M}_{ext} = \begin{pmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}_{1\times 3} & 1 \end{pmatrix} \tag{19}$$

with $\mathbf{R} \in \mathbb{R}^{3\times 3}$ describing the rotation and $\mathbf{t} \in \mathbb{R}^{3\times 1}$ describing the translation of the world coordinate system.

其中$\mathbf{R} \in \mathbb{R}^{3\times 3}$描述旋转，$\mathbf{t} \in \mathbb{R}^{3\times 1}$描述世界坐标系的平移。

With this matrix the homogeneous world coordinates are converted into homogeneous camera coordinates .

通过这个矩阵，齐次世界坐标被转换为齐次相机坐标。

While the camera matrices are commonly available in modern broadcasting cameras [31] , they can also be calculated, for example, by using a Direct Linear Transform [1] .

虽然摄像机矩阵在现代转播摄像机中很常见[31]，但它们也可以通过例如直接线性变换[1]来计算。

This is done particularly straightforward in the context of sports, because characteristic points on the court can be easily annotated automatically.

这在体育运动中尤其简单，因为可以轻松地自动标注球场上的特征点。

In the synthetic dataset the camera matrices are provided by the physics engine MuJoCo, thus, the exact matrices are already available.

在合成数据集中，相机矩阵由物理引擎 MuJoCo 提供，因此，精确的矩阵已经可用。

In the real dataset the camera is calibrated such that intrinsic camera matrix is already known.

在真实数据集中，相机经过标定，使得相机内参矩阵已知。

Therefore, we only have to calculate the extrinsic camera matrix.

因此，我们只需计算相机外参矩阵即可。

We label 11 characteristic points of the scene for each camera location and calculate an initial guess of the extrinsic camera matrix using the Direct Linear Transform.

我们为每个摄像机位置标记场景的 11 个特征点，并使用直接线性变换计算外部摄像机矩阵的初始猜测。

We then further optimize the extrinsic camera matrix using the BFGS optimization algorithm [3] .

然后我们使用BFGS优化算法[3]进一步优化相机外参矩阵。

This way we obtain an accurate estimation of the extrinsic camera matrix.

这样我们就可以获得相机外参矩阵的准确估计。

As this estimation is accurate but not exact, we conclude that our model is able to cope with approximate camera locations which is an important property for practical applications.

由于这种估计是准确的但不精确，我们得出的结论是我们的模型能够处理近似的相机位置，这对于实际应用来说是一个重要的属性。

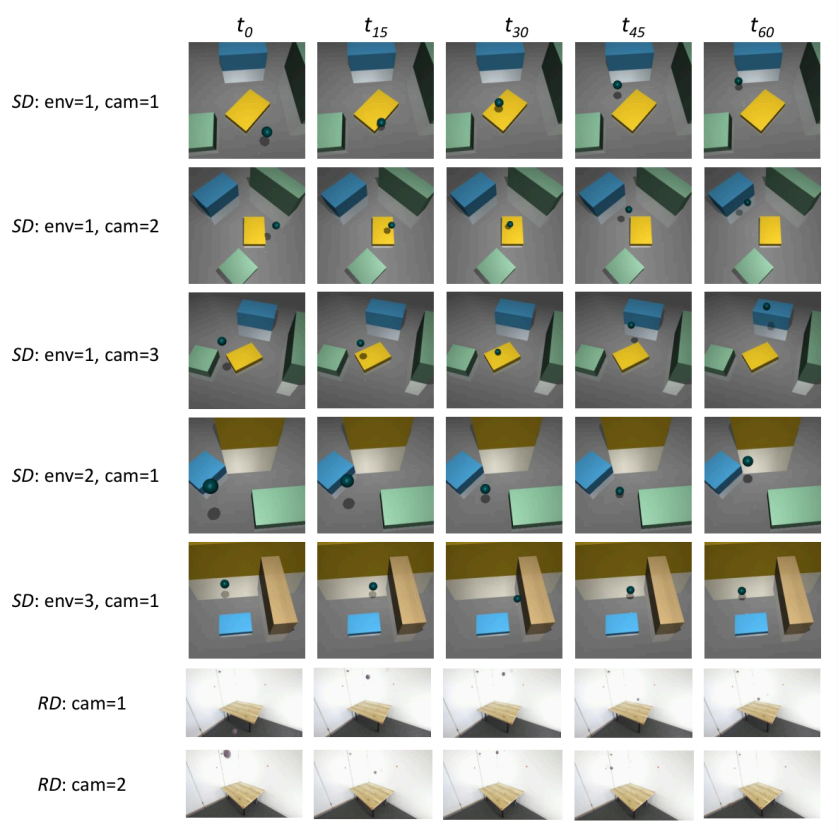


Figure 9 : Frames from example sequences of multiple camera locations and environments are shown.

图 9：显示了来自多个摄像机位置和环境的示例序列的帧。

8 Dataset

8 数据集

In this section, we present additional details on the datasets introduced in the main paper.

在本节中，我们将介绍主论文中介绍的数据集的更多详细信息。

In Figure 9 , we visualize some exemplary videos from different camera positions and in different environments to provide a better overview of the datasets.

在图 9 中，我们可视化了来自不同相机位置和不同环境的一些示例视频，以提供对数据集的更好概述。

In the synthetic dataset , we record the videos from 9 possible camera locations that are shown in Table 8 .

在合成数据集中，我们记录了表 8 中所示的 9 个可能的摄像机位置的视频。

All camera locations are chosen such that the origin of the world coordinate system corresponds to the center of the image.

选择所有相机位置，使世界坐标系的原点对应于图像的中心。

We set the gravitational constant to $g = 1\,\frac{m}{s^2}$.

我们将重力加速度设为 $g = 1\,\frac{m}{s^2}$。

This reduces the speed of the ball, making it easier to automatically generate the synthetic dataset, because we have to ensure that the ball does not leave the field of view.

这降低了球的速度，使得自动生成合成数据集变得更容易，因为我们必须确保球不会离开视野。

Since we record all videos in the real dataset with a ZED 2i Stereo Camera, we are able to calculate a ground truth depthmap for each image.

真实数据集中的视频均由 ZED 2i 双目相机拍摄，因此可以计算每幅图像对应的真值深度图。

For this purpose, we select the Neural Depth Mode of the camera to obtain the best possible results.

为此，我们选择相机的神经深度模式以获得最佳效果。

Nevertheless, the results are still noisy and, thus, we extract the ground truth depth at the annotated ball’ s position as average over the neighboring values of the depth map D as

尽管如此，结果仍然有噪声，因此，我们提取带标注的球位置处的真值深度作为深度图相邻值的平均值D：

$$z^{(C)} = \frac{1}{9} \cdot \sum_{i=-1}^{1} \sum_{j=-1}^{1} D_{x^{(I)}+i,\, y^{(I)}+j} \tag{20}$$

with $\begin{pmatrix} x^{(I)} & y^{(I)} \end{pmatrix}^T$ being the ground truth ball’ s position in image coordinates.

其中$\begin{pmatrix} x^{(I)} & y^{(I)} \end{pmatrix}^T$为图像坐标中球的真值位置。

Based on this ground truth depth and the intrinsic camera matrix we calculate the ground truth camera coordinates.

基于这个真值深度和相机内参矩阵，我们计算真值相机坐标。

We define a fixed origin of the world coordinate system such that the physical potentials can be described easily, and we visualize the origin in Figure 10 .

我们定义了世界坐标系的固定原点，以便可以轻松描述物理势能，并在图 10 中可视化该原点。

The 2D image coordinates are annotated manually by simply clicking on the ball’ s position in the image.

只需单击图像中球的位置即可手动标注 2D 图像坐标。

To expedite this process, we use a simple automatic ball detection algorithm based on the Hough Transformation and only correct the ball position manually if the automatic detection fails.

为了加快这一过程，我们使用基于霍夫变换的简单自动球检测算法，并且仅在自动检测失败时才手动纠正球位置。

camera 相机	d (m)	θ (°)	ϕ (°)
1	8	−60	40
2	10	−65	0
3	10	−55	60
4	9	−60	20
5	7	−70	50
6	9	−50	10
7	10	−60	30
8	8	−70	0
9	8	−50	60

Table 3 : Camera locations in the synthetic dataset.

表 3：合成数据集中的相机位置。

Each camera looks at the origin of the world-coordinate system and is positioned at a radial distance d , polar angle θ and azimuthal angle ϕ with respect to the origin.

每个相机均着眼于世界坐标系的原点，并定位在相对于原点的径向距离d、极角θ和方位角ϕ处。

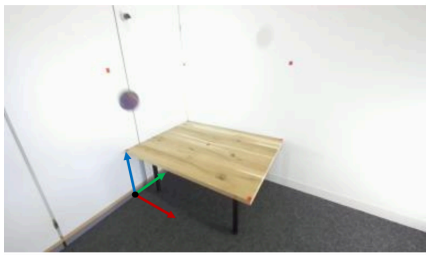


Figure 10 : Origin of the real dataset .

图 10：真实数据集的坐标原点。

Blue represents the z-axis, green the y-axis and red the x-axis.

蓝色代表 z 轴，绿色代表 y 轴，红色代表 x 轴。

9 Experiments

9 实验

In this section we give further details on the training process, provide additional results from the main paper and discuss new experiments.

在本节中，我们将进一步详细介绍训练过程，提供主论文的其他结果并讨论新的实验。

Discussion of training process

训练流程讨论

Our code is implemented in PyTorch [24] , and we utilize models and architectures from the timm library [33] .

我们的代码是在 PyTorch [24] 中实现的，并且我们利用 timm 库 [33] 中的模型和架构。

We optimize our model with the ADAM [15] optimizer and use a batch size of 8 together with a learning rate of $2 \cdot 10^{-5}$.

我们使用 ADAM [15] 优化器优化模型，并使用8的批量大小和$2 \cdot 10^{-5}$的学习率。

The Dormand-Prince method [8] is applied for the numerical solution of the differential equations 5 and its implementation utilizes the torchdyn [25] library.

Dormand-Prince 方法 [8] 用于微分方程 5 的数值求解，其实现利用 torchdyn [25] 库。

We train the RD and SD-S models for 400 epochs, the SD-M model for 200 epochs and the SD-L model for 100 epochs.

我们训练 RD 和 SD-S 模型 400 个 epoch，SD-M 模型训练 200 个 epoch，SD-L 模型训练 100 个 epoch。

Although the models show slight improvements with longer training, the performance gains are small.

尽管模型在较长时间的训练中表现出轻微的改进，但性能提升很小。

We evaluate the model after each epoch on the validation set and select the model with the best performance for the final evaluation on the test set.

我们在验证集上的每个时期之后评估模型，并选择性能最佳的模型用于测试集上的最终评估。

We multiply the future loss (first term in equation 9) with a factor, which linearly increases from 0 to 1 during the first 300 epochs of the training.

我们将未来预测损失（等式 9 中的第一项）乘以一个因子，该因子在训练的前 300 个时期内从 0 线性增加到 1。

This way we ensure that the model focuses on learning to recognize the ball in the images first, and thus, prevents a model collapse.

通过这种方式，我们确保模型首先专注于学习识别图像中的球，从而防止模型崩溃。

Our experiments were conducted on a combination of Nvidia RTX 3090, V100, A100, and H100 GPUs.

我们的实验是在 Nvidia RTX 3090、V100、A100 和 H100 GPU 的组合上进行的。

Therefore, we do not provide the actual runtime of our experiments as a comparison is not meaningful.

因此，我们不提供实验的实际运行时间，因为比较没有意义。

However, to provide a general idea of the runtime, training our model for 400 epochs on the real dataset in Table 2 takes approximately 12 hours on a single V100 GPU.

然而，为了提供运行时的总体概念，在表 2 中的真实数据集上训练我们的模型 400 个周期，在单个 V100 GPU 上大约需要 12 小时。

We note that a considerable good performance is already reached after only 100 epochs (3 hours on a single V100 GPU).

我们注意到，仅 100 个 epoch（单个 V100 GPU 上 3 小时）后就已经达到了相当好的性能。

Furthermore, we anticipate that further code optimizations, such as improved data loading techniques or model compilation, could lead to significant increases in training speed.

此外，我们预计进一步的代码优化，例如改进的数据加载技术或模型编译，可能会导致训练速度的显著提高。

For all results in this paper we report the mean and standard deviation (represented by the symbol Δ) of the metric calculated across all images in the test set.

对于本文中的所有结果，我们报告了测试集中所有图像计算得出的指标的平均值和标准差（由符号Δ表示）。

While we provide only results for camera locations 1, 7, 8, and 9 in Table 1(a) , the results for all camera locations are given in Table 4 .

虽然我们在表 1(a) 中仅提供了摄像机位置 1、7、8 和 9 的结果，但表 4 中给出了所有摄像机位置的结果。

We see that the SD-M and SD-L models are able to estimate the ball’ s 3D position for all camera locations very precisely.

我们看到 SD-M 和 SD-L 模型能够非常精确地估计所有摄像机位置的球的 3D 位置。

Since the SD-L model achieves better scores for most camera locations, this reinforces our assumption that training with additional data recorded in multiple environments is actually beneficial.

由于 SD-L 模型在大多数相机位置上获得了更好的分数，这强化了我们的假设，即使用在多个环境中记录的附加数据进行训练实际上是有益的。

Table 4: DtG scores per camera location evaluated on the 1 st environment of the synthetic dataset .

表 4：在合成数据集的第一个环境上评估的每个相机位置的 DtG 分数。

An extension of Table 1(a) from the main paper.

表 1(a) 来自主论文的扩展。

	DtG ± ΔDtG (cm) ↓								
training set **训练集**	camera 1 相机1	camera 2 相机2	camera 3 相机3	camera 4 相机4	camera 5 相机5	camera 6 相机6	camera 7 相机7	camera 8 相机8	camera 9 相机9
SD-S **SD-S**	22 ± 19	-	-	-	-	-	-	-	-
SD-M **SD-M**	19 ± 10	19 ± 18	28 ± 24	13 ± 9	26 ± 12	19 ± 14	27 ± 23	23 ± 9	21 ± 10
SD-L **SD-L**	11 ± 6	20 ± 19	28 ± 25	11 ± 8	10 ± 6	19 ± 13	28 ± 25	15 ± 8	16 ± 7

Comparison of different backbones

不同主干网络的比较

We perform an additional experiment to compare the performance of different backbones in the PEN.

我们进行了一项额外的实验来比较 PEN 中不同主干网的性能。

For each backbone, we use the implementation provided in the timm library.

对于每个主干网络，我们使用 timm 库中提供的实现。

In the other experiments, we always use the first 4 out of 5 stages of a ResNet34.

在其他实验中，我们始终使用 ResNet34 5 个阶段中的前 4 个阶段。

We now compare this backbone with 4 out of 4 stages of the ConvNeXt-nano architecture and 4 out of 4 stages of the ConvNeXtv2-nano architecture.

现在，我们将此主干与 ConvNeXt-nano 架构的 4 个阶段中的 4 个以及 ConvNeXtv2-nano 架构的 4 个阶段中的 4 个进行比较。

All models are pretrained on ImageNet-1K, and we train and test them on the SD-M dataset.

所有模型都在 ImageNet-1K 上进行预训练，我们在 SD-M 数据集上训练和测试它们。

The results are shown in table 5 .

结果如表5所示。

Table 5: Results for different backbones.

表 5：不同主干网络的结果。

Each model is trained on the SD-M dataset.

每个模型都在 SD-M 数据集上进行训练。

DtG scores per camera location evaluated on the 1 st environment of the synthetic dataset .

在合成数据集的第一个环境上评估每个摄像机位置的 DtG 分数。

		DtG ± ΔDtG (cm) ↓								
backbone 主干网络	# params # 参数	camera 1 相机1	camera 2 相机2	camera 3 相机3	camera 4 相机4	camera 5 相机5	camera 6 相机6	camera 7 相机7	camera 8 相机8	camera 9 相机9
resnet ResNet	$0.8 \cdot 10^7$	19 ± 10	19 ± 18	28 ± 24	13 ± 9	26 ± 12	19 ± 14	27 ± 23	23 ± 9	21 ± 10
convnext ConvNeXt	$1.5 \cdot 10^7$	14 ± 12	26 ± 21	32 ± 27	15 ± 13	15 ± 11	24 ± 21	37 ± 31	22 ± 11	20 ± 12
convnextv2 ConvNeXtv2	$1.5 \cdot 10^7$	23 ± 14	33 ± 28	45 ± 32	25 ± 20	18 ± 11	31 ± 27	49 ± 36	53 ± 22	35 ± 20

Interestingly, the ConvNeXtv2 backbone performs notably worse than the other two architectures, despite its seemingly advanced design.

有趣的是，尽管 ConvNeXtv2 主干网的设计看似先进，但其性能明显比其他两种架构差。

In summary, the ResNet performs slightly better than the ConvNeXt, despite the ConvNeXt having twice as many parameters.

总体而言，虽然 ConvNeXt 的参数量是 ResNet 的 2 倍，ResNet 的表现仍略好一些。

One reason for this behavior might be that our input image resolution is small, and we do not benefit from a very large receptive field.

造成这种行为的原因之一可能是我们的输入图像分辨率很小，并且我们无法从非常大的感受野中受益。

Since the ConvNeXt architecture uses 7×7 convolutions, the receptive field grows faster than in the ResNet.

由于 ConvNeXt 架构使用7×7卷积，因此感受野比 ResNet 增长得更快。

Once the receptive field size becomes as large as the image resolution, the additional stages may not provide significant benefits and could potentially hinder the model’ s ability to learn an effective representation.

一旦感受野大小变得与图像分辨率一样大，额外的阶段可能不会提供显著的好处，并且可能会阻碍模型学习有效表示的能力。

Since the ResNet only uses 3×3 convolutions and the features are extracted at an earlier stage compared to the other architectures, the receptive field is smaller and the model might be able to learn a better representation.

由于 ResNet 仅使用3×3卷积，并且与其他架构相比，特征是在较早阶段提取的，因此感受野更小，模型可能能够学习更好的表示。

However, additional experiments using different backbones and higher input image resolutions are required to validate this hypothesis.

然而，需要使用不同主干和更高输入图像分辨率的额外实验来验证这一假设。

We note that our code can be readily extended with other backbones, provided that they support the extraction of 4 feature stages, which is a requirement in our implementation.

我们注意到，我们的代码可以很容易地用其他主干进行扩展，只要它们支持 4 个特征阶段的提取，这是我们实现中的要求。

If a different number of features is used, the implementation of the depthmap head and heatmap head needs to be adjusted.

如果使用不同数量的特征，则需要调整深度图头和热图头的实现。

In conclusion, even though the ResNet is a relatively old architecture, it remains a highly suitable choice for our task.

总之，尽管 ResNet 是一个相对较旧的架构，但它仍然是非常适合我们任务的选择。

Analysis of future loss

未来预测损失分析

In this section, we delve deeper into the concept of the future loss .

在本节中，我们将深入探讨未来预测损失的概念。

As can be seen in figure 2 , we compare the image coordinates $\vec{\mathbf{r}}^{(I)}_{\mathrm{PEN}}(t_n+\Delta n)$ predicted by the PEN with the coordinates $\vec{\mathbf{r}}^{(I)}_{\mathrm{PAF}}(t_n+\Delta n)$ forecasted by the PAF.

如图2所示，我们将PEN预测的图像坐标$\vec{\mathbf{r}}^{(I)}_{\mathrm{PEN}}(t_n+\Delta n)$与PAF预测的图像坐标$\vec{\mathbf{r}}^{(I)}_{\mathrm{PAF}}(t_n+\Delta n)$进行比较。

This way of computing the future loss is explicitly written in the first term of equation 9 , and we refer to this method as 2D-predict in this section.

这种计算未来预测损失的方法明确地写在等式 9 的第一项中，我们在本节中将该方法称为 2D 预测。

Because the ground truth image coordinates are available for every frame, we can also compute the future loss by comparing the PEN’ s predictions with the ground truth coordinates $\vec{\mathbf{r}}^{(I)}_{\mathrm{GT}}(t_n+\Delta n)$.

由于每一帧都有真值图像坐标，我们还可以通过将 PEN 的预测与真值坐标$\vec{\mathbf{r}}^{(I)}_{\mathrm{GT}}(t_n+\Delta n)$进行比较来计算未来预测损失。

Consequently, we simply replace the coordinates $\vec{\mathbf{r}}^{(I)}_{\mathrm{PAF}}(t_n+\Delta n)$ with $\vec{\mathbf{r}}^{(I)}_{\mathrm{GT}}(t_n+\Delta n)$ in equation 9 .

因此，我们只需将式9中的坐标$\vec{\mathbf{r}}^{(I)}_{\mathrm{PAF}}(t_n+\Delta n)$替换为$\vec{\mathbf{r}}^{(I)}_{\mathrm{GT}}(t_n+\Delta n)$即可。

This way of computing the loss is denoted as 2D-gt .

这种计算损失的方法被表示为 2D-gt 。

Moreover, we also add a third method of computing the future loss by comparing the PEN’ s predictions with the ground truth camera coordinates $\vec{\mathbf{r}}^{(C)}_{\mathrm{GT}}(t_n+\Delta n)$, which are usually not used for the training.

此外，我们还添加了第三种计算未来预测损失的方法，即通过将 PEN 的预测与真值相机坐标$\vec{\mathbf{r}}^{(C)}_{\mathrm{GT}}(t_n+\Delta n)$进行比较，该坐标通常不用于训练。

Consequently, the future loss is computed in this case as

因此，在这种情况下，未来预测损失计算为

$$\text{future loss} = \left\| \vec{\mathbf{r}}^{(C)}_{\mathrm{PEN}}(t_n+\Delta n) - \vec{\mathbf{r}}^{(C)}_{\mathrm{GT}}(t_n+\Delta n) \right\|_{\mathrm{L1}} . \tag{21}$$

We refer to this way of computing the loss as 3D-gt in this section, and we use this method as baseline.

在本节中，我们将这种计算损失的方法称为 3D-gt，并使用该方法作为基线。

We note that for this way of computing the loss, we utilize the 3D ground truth, which is usually not available in real world scenarios.

我们注意到，对于这种计算损失的方式，我们利用 3D 真值，这在现实世界场景中通常不可用。

Nonetheless, we believe that examining this baseline can provide valuable insights into our analysis.

尽管如此，我们相信检查这一基线可以为我们的分析提供有价值的见解。

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Table 6: DtG scores and DtG_3 scores per camera location for the real dataset .

表 6：真实数据集的每个相机位置的 DtG 分数和DtG_3分数。

Models trained with different future losses are compared.

比较使用不同未来预测损失训练的模型。

	$\mathrm{DtG} \pm \Delta\mathrm{DtG}$ **(cm)** ↓		$\mathrm{DtG}_3 \pm \Delta\mathrm{DtG}_3$ **(cm)** ↓	
future loss **未来预测损失**	**camera 1** **相机1**	**camera 2** **相机2**	**camera 1** **相机1**	**camera 2** **相机2**
2D-predict **二维预测**	7 ± 4	6 ± 4	$\begin{pmatrix} 8 \pm 4 \\ 6 \pm 2 \\ 11 \pm 4 \end{pmatrix}$	$\begin{pmatrix} 4 \pm 2 \\ 6 \pm 3 \\ 11 \pm 4 \end{pmatrix}$
2D-gt **2D-gt**	18 ± 10	17 ± 10	$\begin{pmatrix} 6 \pm 5 \\ 14 \pm 7 \\ 33 \pm 7 \end{pmatrix}$	$\begin{pmatrix} 5 \pm 3 \\ 15 \pm 6 \\ 32 \pm 7 \end{pmatrix}$
3D-gt **3D-gt**	3 ± 3	3 ± 4	$\begin{pmatrix} 3 \pm 2 \\ 2 \pm 2 \\ 6 \pm 5 \end{pmatrix}$	$\begin{pmatrix} 3 \pm 2 \\ 2 \pm 3 \\ 6 \pm 5 \end{pmatrix}$

We train three models with the different future losses and report the results in table 6 .

我们训练了具有不同未来预测损失的三个模型，并在表 6 中报告了结果。

By comparing the 2D-predict model with the 2D-gt model, we can see that the 2D-predict model performs better than the 2D-gt model.

通过将 2D-predict 模型与 2D-gt 模型进行比较，我们可以看到 2D-predict 模型比 2D-gt 模型表现更好。

This is due to the ground truth image coordinates being noisy, since they are annotated manually.

这是由于真值图像坐标有噪声，因为它们是手动标注的。

Obviously, the PEN learns to predict more accurate image coordinates and, consequently, the calculation of the future loss is more accurate.

显然，PEN 学会预测更准确的图像坐标，因此，未来预测损失的计算也更准确。

As expected, the 3D-gt model performs best, since it is trained with 3D ground truth information.

正如预期的那样，3D-gt 模型表现最好，因为它是使用 3D 真值信息进行训练的。

However, despite the 2D-predict model being worse, it is still able to predict the 3D position of the ball surprisingly well, since the metric is in the same order of magnitude as with the 3D-gt model.

然而，尽管 2D 预测模型较差，但它仍然能够非常好地预测球的 3D 位置，因为该指标与 3D-gt 模型处于同一数量级。

Consequently, we conclude that the method presented in this paper provides a viable alternative to fully supervised training.

因此，我们得出的结论是，本文提出的方法为完全监督训练提供了可行的替代方案。

Additionally, we highlight the potential for combining our method with supervised training, particularly in scenarios where 3D ground truth data is only accessible for a subset of the data.

此外，我们强调了将我们的方法与监督训练相结合的势能，特别是在仅可访问数据子集的 3D 真值数据的情况下。

Such integration could lead to further enhancements in existing applications.

这种集成可以进一步增强现有应用程序。

Learning arbitrary physics

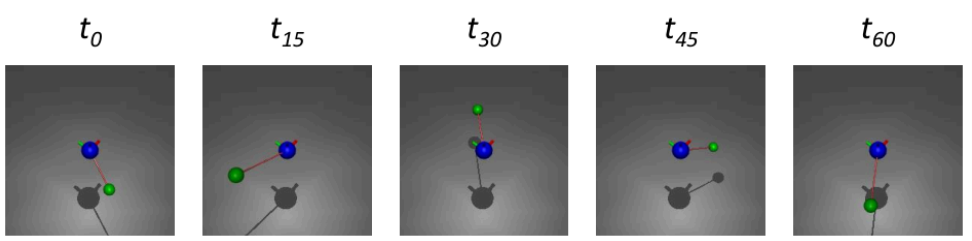


Figure 11 : Frames from an example video of the spring dataset .

图 11：来自 弹簧数据集示例视频的帧。

An important advantage of our method is its ability to describe arbitrary physical systems by utilizing numerical solutions for the equations of motion, rather than relying on analytical solutions.

我们的方法的一个重要优点是它能够通过利用运动方程的数值解来描述任意物理系统，而不是依赖于解析解。

In this section we show two things:

在本节中，我们展示两件事：

- We demonstrate that our method can also be used for physical systems that behave differently from the previous described bouncing balls.

  我们证明我们的方法也可以用于行为与之前描述的弹跳球不同的物理系统。

- We calculate an analytic solution of the equations of motion and compare the numerical forecasting in the PAF with the analytic forecasting.

  我们计算运动方程的解析解，并将 PAF 中的数值预测与解析预测进行比较。

Therefore, we introduce a new synthetic dataset denoted as spring dataset .

因此，我们引入了一个新的合成数据集，表示为 spring dataset 。

It consists of video of a moving ball connected with a spring to a fixed ball in the center.

它由一个通过弹簧连接到中心固定球的移动球的视频组成。

The moving ball of mass $m = 1\ \mathrm{kg}$ is attracted to the fixed center ball by the spring with spring constant $k = 3\ \frac{N}{m}$.

质量$m = 1\ \mathrm{kg}$的移动球被弹簧常数$k = 3\ \frac{N}{m}$的弹簧吸引到固定中心球上。

We record 100 videos from a single fixed camera location for training, 100 videos for validation and 100 videos for testing.

我们从单个固定摄像机位置录制 100 个视频用于训练，100 个视频用于验证，100 个视频用于测试。

The resolution of the frames is 224×224 and the videos are recorded with $30\ \mathrm{FPS}$.

帧的分辨率为224×224，视频录制为$30\ \mathrm{FPS}$。

We visualize frames from an example video depicting the motion of the ball connected to the spring in Figure 11 .

我们将示例视频中的帧可视化，该视频描述了连接到弹簧的球的运动，如图 11 所示。

According to Hooke’ s law, the potential of the moving ball is given by $V(\vec{\mathbf{r}}^{(W)}) = \frac{1}{2}k\left|\vec{\mathbf{r}}^{(W)}\right|^2$ and by applying the Hamilton formalism (see equation 5), we obtain the equations of motion as

根据胡克定律，运动球的势由$V(\vec{\mathbf{r}}^{(W)}) = \frac{1}{2}k\left|\vec{\mathbf{r}}^{(W)}\right|^2$给出，并应用哈密顿形式（见方程5），我们得到运动方程为

$$\frac{d}{\mathrm{dt}}\vec{\mathbf{r}}^{(W)} = \vec{\mathbf{v}}, \frac{d}{\mathrm{dt}}\vec{\mathbf{v}} = -\frac{k}{m}\cdot\vec{\mathbf{r}}^{(W)}. \tag{22}$$

The analytic solution to these differential equations can be experessed as

这些微分方程的解析解可以表示为

$$\vec{\mathbf{r}}^{(W)}(t) = \frac{1}{k}\cdot\vec{\mathbf{v}}_0\cdot\sin(k\cdot t) + \vec{\mathbf{r}}_0^{(W)} \tag{23}$$

where $\vec{\mathbf{r}}_0^{(W)}$ and $\vec{\mathbf{v}}_0$ are the initial position and velocity of the moving ball.

其中$\vec{\mathbf{r}}_0^{(W)}$和$\vec{\mathbf{v}}_0$是移动球的初始位置和速度。

We implement the numeric solver of equation 22 similar to the previous experiments, and compare its performance with the utilization of the analytic function 23 in the PAF.

我们实现了类似于之前实验的方程 22 的数值求解器，并将其性能与 PAF 中解析函数 23 的使用进行了比较。

The results are given in Table 7 .

结果如表7所示。

Table 7: DtG and DtG_3 scores on the spring dataset .

表 7：弹簧数据集上的 DtG 和DtG_3得分。

The analytic solution of the equations of motion is compared to a numeric solution.

将运动方程的解析解与数值解进行比较。

PAF mode PAF模式	$\mathrm{DtG} \pm \Delta\mathrm{DtG}$ (cm) ↓	$\mathrm{DtG}_3 \pm \Delta\mathrm{DtG}_3$ (cm) ↓
numeric 数值求解	9 ± 15	$\begin{pmatrix} 6 \pm 6 \\ 5 \pm 6 \\ 16 \pm 22 \end{pmatrix}$
analytic 解析求解	10 ± 16	$\begin{pmatrix} 8 \pm 8 \\ 4 \pm 5 \\ 21 \pm 21 \end{pmatrix}$

The model trained with the numeric PAF and the model trained with the analytical PAF perform very similarly, with the numeric version achieving slightly better scores.

采用数值 PAF 和解析 PAF 训练的两个模型表现相近，其中数值版本略优。

This shows that using numeric solvers in the PAF is a valid approach that is able to describe arbitrary physical systems.

这表明在 PAF 中使用数值求解器是一种能够描述任意物理系统的有效方法。

In general, the numeric model achieves a DtG of just $9\ \mathrm{cm}$, demonstrating the applicability of our method to physical systems beyond bouncing balls.

一般来说，数值模型的 DtG 仅为$9\ \mathrm{cm}$，证明了我们的方法对弹跳球之外的物理系统的适用性。

While the physics of all datasets presented in this paper is determined by the laws of classical mechanics, it is worth considering the potential extension of our method to other domains such as quantum mechanics or electrodynamics, where analytic solutions are not available in most cases.

虽然本文中提出的所有数据集的物理原理都是由经典力学定律决定的，但值得考虑我们的方法可能扩展到其他领域，例如量子力学或电动力学，在大多数情况下，解析解不可用。

However, exploring these possibilities lies beyond the scope of this work.

然而，探索这些可能性超出了这项工作的范围。

Discussion of stability

稳定性讨论

To ensure that our results are not just random fluctuations, we test our model with different random seeds and compare the results.

为了确保我们的结果不仅仅是随机波动，我们使用不同的随机种子测试我们的模型并比较结果。

For this experiment we choose the SD-S scenario from the main paper and set 8 different random seeds at the beginning of the training.

对于这个实验，我们选择主论文中的 SD-S 场景，并在训练开始时设置 8 个不同的随机种子。

This results in a different initialization of the heatmap head and depthmap head , as well as in a different ordering of the training data in each epoch.

这导致热图 head 和深度图 head 的初始化不同，以及每个时期训练数据的不同排序。

We present the results of the models in Table 8 .

我们在表 8 中列出了模型的结果。

Table 8: DtG scores for the 1 st camera location in the 1 st environment.

表 8：第一个环境中第一个摄像机位置的 DtG 分数。

The results of the SD-S model are compared for different initial seeds.

对不同初始种子的 SD-S 模型结果进行了比较。

seed 种子	$\mathrm{DtG} \pm \Delta\mathrm{DtG}$ **(cm)** ↓
1	24 ± 20
2	27 ± 21
3	26 ± 21
4	23 ± 19
5	23 ± 21
6	25 ± 21
7	25 ± 19
8	22 ± 21

The results obtained for all seed values show a high degree of similarity.

所有种子值获得的结果显示出高度的相似性。

The mean value across multiple runs is $E_{\mathrm{DtG}} = 24.4\ \mathrm{cm}$, with a small standard deviation of $\sigma_{\mathrm{DtG}} = 1.6\ \mathrm{cm}$.

多次运行的平均值为$E_{\mathrm{DtG}} = 24.4\ \mathrm{cm}$，较小的标准差为$\sigma_{\mathrm{DtG}} = 1.6\ \mathrm{cm}$。

The low standard deviation among individual runs suggests that the outcomes are not mere random fluctuations.

各个运行之间的低标准差表明结果不仅仅是随机波动。

This observation is further supported by the fact that the standard deviation over the multiple runs is an order of magnitude smaller than the standard deviations among the individual images $\Delta\mathrm{DtG}$.

多次运行的标准差比单个图像之间的标准差小一个数量级$\Delta\mathrm{DtG}$，这一事实进一步支持了这一观察结果。

Hence, the effect of the random seed is negligible.

因此，随机种子的影响可以忽略不计。

While the results are relatively stable over different seeds, we observe that other factors have a much larger impact on the results.

虽然不同种子的结果相对稳定，但我们观察到其他因素对结果有更大的影响。

One such factor is the selection of the number of forecast steps.

这样的因素之一是预测步骤数量的选择。

As described in equation 9 , we compare the coordinates at time $t_{n+\Delta n}$ with the forecasted coordinates at this time.

如式9所示，我们将$t_{n+\Delta n}$时刻的坐标与此时的预测坐标进行比较。

Depending on the speed of the ball, a good value for Δn has to be chosen carefully.

根据球的速度，必须仔细选择Δn的良好值。

If Δn is too small, the ball may not have moved a significant distance, resulting in a very low loss.

如果Δn太小，球可能不会移动很远的距离，导致损失非常低。

Consequently, the PEN may converge to a trivial solution.

因此，PEN 可能收敛到平凡解。

Conversely, if Δn is too large, the distance might be too large, and the PEN may struggle to learn the correct solution.

相反，如果Δn太大，距离可能太大，PEN 可能很难学习正确的解决方案。

Therefore, for each scenario an appropriate value for Δn has to be chosen.

因此，对于每种情况，必须为Δn选择适当的值。

In this paper, we do not use a single value for Δn , instead we calculate the future loss for multiple values of Δn and then compute the average.

在本文中，我们不使用Δn的单个值，而是计算Δn的多个值的未来预测损失，然后计算平均值。

This way, the small Δn values help to stabilize the training, while larger ones ensure that the PEN does not collapse to a trivial solution.

较小的 Δn 有助于稳定训练，而较大的取值能防止 PEN 退化为平凡解。

Although this approach aids in training, it is still necessary to choose appropriate values for Δn .

虽然这种方法有助于训练，但仍然需要为Δn选择合适的值。

One way to significantly improve the stability of the training is to use a better pretraining of the PEN.

显著提高训练稳定性的一种方法是使用更好的 PEN 预训练。

For example a segmentation or even a depth estimation task could be chosen for pretraining the model.

例如，可以选择分割甚至深度估计任务来预训练模型。

Since this teaches the model 3D knowledge, a model collapse is less likely to happen.

由于这会教授模型 3D 知识，因此模型崩溃的可能性较小。

However, in this paper we want to show that our method is able to learn 3D dynamics without any 3D supervision.

不过，本文希望验证的是：该方法能够在完全没有三维监督的情况下学习三维动力学。

Thus, we do not further explore more advanced pretraining strategies, and instead only initialize the backbone of the PEN with ImageNet weights in our experiments.

因此，我们没有进一步探索更先进的预训练策略，而是在实验中仅使用 ImageNet 权重初始化 PEN 的主干。

Limitations and improvements

限制和改进

We view this paper as a step towards 3D object location estimation without requiring 3D supervision.

我们将本文视为无需 3D 监督即可进行 三维目标位置估计的一步。

Our findings demonstrate the feasibility of training a model without depth information and provide a comprehensive analysis of our approach.

我们的研究结果证明了在没有深度信息的情况下训练模型的可行性，并对我们的方法进行了全面分析。

However, there are still several limitations to address.

然而，仍有一些限制需要解决。

Specifically, we only use low resolution images to ensure an efficient training of our models.

具体来说，我们只使用低分辨率图像来确保模型的有效训练。

It is possible to use such low resolution images in our experiments, because the ball’ s relative size in the images is large enough to be clearly identified.

在我们的实验中可以使用如此低分辨率的图像，因为图像中球的相对尺寸足够大，可以清楚地识别。

In contrast, sports videos typically involve high-resolution frames where the ball appears much smaller.

相比之下，体育视频通常涉及高分辨率帧，其中球显得小得多。

Consequently, simply downscaling the entire image would not suffice to maintain clear ball identification.

因此，简单地缩小整个图像的比例不足以保持清晰的球识别。

Hence, the development of more sophisticated methods is necessary to optimize computation time.

因此，有必要开发更复杂的方法来优化计算时间。

One approach is to leverage an object detection pipeline to extract the relevant region around the ball, which can then be utilized for 3D location estimation.

一种方法是利用对象检测流程来提取球周围的相关区域，然后将其用于 3D 位置估计。

We anticipate that the insights presented in this paper will inspire the advancement of more sophisticated techniques in this field.

我们预计本文提出的见解将激发该领域更复杂技术的进步。